EMD51806 PROJECT PHASE-1

Cognitive Socially Assistive Robot for Multimodal Assessment and Adaptive Support in Autism Therapy

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IN

MECHATRONICS

HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
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DEPARTMENT OF MECHATRONICS ENGINEERING

HINDUSTAN INSTITUTE OF TECHNOLOGY AND SCIENCE

NOVEMBER 2025

BONAFIDE CERTIFICATE

Certified that this project report titled **"Cognitive socially assistive robot for multimodal assessment and adaptive support in autism therapy"** is the Bonafide work of

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ABSTRACT

Providing intervention for individuals with autism spectrum disorder (ASD) and social communication deficits, joint attention weaknesses, and emotional dysregulation necessitates ongoing, individualized, and data-informed interventions; areas where evidence of efficacy in traditional human-delivered clinical models is limited due to scalability, objectivity, and replicability. The current study proposes the Cognitive Socially Assistive Robot (C-SAR), a mobile and intelligent robot system created to assist in increasing therapeutic engagement and social learning for children with ASD. C-SAR is implemented on a TurtleBot3 mobile robot platform, which was chosen due to its modular design, compatibility with ROS (Robot Operating System), and appropriateness for therapy with human-robot interaction. The TurtleBot3 acts as the driving force behind C-SAR, affording safe autonomous navigation, adaptive movement, and real-time interaction in a controlled setting. C-SAR integrates assessment modules regarding multimodal behavior such as eye gaze tracking, facial expression recognition, and speech analysis to create three behavioral indices, Joint Attention Score (JAS), Social Engagement Score (SES), and Speech Clarity Score (SCS). During the interactive therapy sessions with the robot, children engaged in adaptive game-based activities with audiovisual feedback. The robot also provided positive reinforcement using expressions, voice modulation, and movements to encourage imitation, attention, and social reciprocity. C-SAR autonomously adapts to the prompts, and can achieve real-time multimodal data processing, using on-device deep learning inference based on the behavioral state of the child. C-SAR also has a therapist dashboard that enables longitudinal data visualization for progress monitoring and treatment of personalizing therapy. Preliminary testing shows promise, with 78–85% for speech imitation accuracy, 85–90% for gaze fixation success, and increased vocal initiation and sustained engagement compared to passive observation tasks. This work offers a mobile, emotionally intelligent, and clinically interpretable robotic solution that enhances traditional therapy models, which is both replicable and scalable for improving social communication skills in children with ASD.

Keywords : Autism Spectrum Disorder (ASD), Socially Assistive Robot, TurtleBot3, Eye Gaze Tracking, Speech Analysis, Human–Robot Interaction

ACKNOWLEDGEMENT

First and foremost, we would like to thank the Lord Almighty for his presence and immense blessings throughout the project work. It’s a matter of pride and privilege to express my deep gratitude to the HITS management for providing me with the necessary facilities and support.

We are highly elated to express my sincere and abundant respect to the Vice-Chancellor , **Dr. S. N. Sridhara**, for giving me this opportunity to present and implement my ideas in this project. We wish to express my heartfelt gratitude to **Dr. D Devika,** Associate Professor and HoDi/c, Department of Mechatronics Engineering, for his valuable support and encouragement in carrying out this work.

We would like to express our sincere gratitude to our guide, **Dr. S. Janaki Raman**, Associate Professor, Department of Mechatronics Engineering, for continually guiding and mentoring us by providing valuable suggestions and insights throughout the completion of our project work.

We would also like to extend our heartfelt thanks to our co-guide, **Ms. N. Seenu**, for her constant encouragement, support, and constructive feedback, which greatly contributed to the successful completion of this project.

We are exceptionally obligated to our project coordinators, **Dr.Sivakamasudari** for their direction and steady supervision throughout our project. We would like to thank all the technical and teaching staff, and the Department of Mechatronics Engineering for all their support.

Last, but not least, my thanks and thanks additionally go to my partner in building up the venture and individuals who have energetically gotten me out with their capacities. We are deeply indebted to my parents who have been the greatest support throughout the project.

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DEPARTMENT OF MECHATRONICS ENGINEER

EMD51512-AI&ROBOTICS

Project Batch No	**:**	10
Title of The Project	**:**	**Cognitive socially assistive robot for multimodal assessment and adaptive support in autism therapy**
Number of The Students In The Batch	**:**	3

Student's Individual Contribution

Name of the Student: Shaik Sameer Salahuddin

Register No.:22125004

Contribution: Integrated the hardware components including the camera, microphone, speaker, and display with the TurtleBot3 platform, implemented ROS2-based teleoperation, and 3D printed the display mount for stable hardware setup.

I was involved in hardware integration, connecting and testing the camera, mic, speaker, and display with the TurtleBot3 robot for seamless operation within the ROS2 framework. I also designed and 3D printed the display mount, ensuring proper alignment and secure attachment during robot-assisted sessions and teleoperation tasks.

Shaik Sameer Salahuddin (22125004)

DEPARTMENT OF MECHATRONICS ENGINEERING

EMD51512-AI&ROBOTICS

Project Batch No	**:**	10
Title of The Project	**:**	**Cognitive socially assistive robot for multimodal assessment and adaptive support in autism therapy**
Number of The Students In The Batch	**:**	3

Student'sIndividualContribution

Nameof the Student: Thirisha. R

Register No.:22125022

Contribution: Developed customized interactive games for children with Autism Spectrum Disorder (ASD) aimed at improving Joint Attention Score (JAS), Social Engagement Score (SES), and Speech Clarity Score (SCS) through multimodal interaction and feedback.

I designed and developed therapeutic games to enhance attention, engagement, and communication in children with ASD. The games were programmed to evaluate and reinforce JAS, SES, and SCS using real-time interaction data, contributing to the behavioral assessment and therapeutic outcomes of the system.

Thirisha.R (22125022)

DEPARTMENT OF MECHATRONICS ENGINEERING

EMD51512-AI&ROBOTICS

Project Batch No	**:**	10
Title of The Project	**:**	**Cognitive socially assistive robot for multimodal assessment and adaptive support in autism therapy**
Number of The Students In The Batch	**:**	3

Student's Individual Contribution

Nameof the Student: AdamHamdullah Muneer Vm

Register No.:22125028

Contribution : Conducted a detailed literature review on socially assistive robotics and analyzed previous research on therapeutic interventions, behavioral metrics, and robot–child interaction models.

I was responsible for reviewing existing research papers and related works in socially assistive robotics, identifying gaps in current therapeutic systems, and correlating findings with the objectives of this project. I also contributed to the final project result analysis and documentation, ensuring that the outcomes align with research benchmarks.

Adam Hamdullah Muneer Vm (2212502

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CHAPTER 1

INTRODUCTION

Autism Spectrum Disorder (ASD) is an increasingly identified neurodevelopmental disability characterized by persistent deficits in social communication, deficits in joint attention, and a pattern of behavior that is restricted or repetitive. The prevalence of ASD is around one in 100 children globally, and the degree of heterogeneity across children with ASD - to measurable outcomes of mild difficulties and verbal expression, to documented disabilities in intellectual and emotional processing is marked. This variation in behaviors supports the notion that evidence-based practices should be utilized with individualization; Children with ASD typically require systematic routine, structured practice and explicit feedback, and a slow and steady approach to reinforce new social and communication skills [1]. Evidence-based practices generally include early intensive behavioural approaches for children with ASD, as these have demonstrated improvements in communicative opportunities, language, emotional understanding and adaptability outcomes for children with ASD [2]. However, these services can be resource intense to execute, particularly with the requirement for on-going service delivery by therapists, suggesting limited scalability of service delivery, especially in rural areas and low-resourced settings.

In many traditional therapy contexts, the evaluation of progress is heavily dependent on the therapist's subjective observations of the children's behaviour; the dependence on the therapist creates some degree of bias and/or inconsistency of measurements. As a result of human nature, some level of human emotional tone, attention control, and fatigue can affect the congruency and quality of an interaction with respect to performance and predictability. Additionally, children with ASD generally are not generalizing newly learned skills/behaviours outside of the controlled clinical context of therapy; and, this limits the effect of intervention over the long-term. Consequently, as a result of these challenges related to intervention, there is more focus on the potential for technology-mediated delivery of therapy for children with ASD, where artificial intelligence (AI), computer vision, and socially assistive robotics contribute structure, objectivity, and adaptability to the behavioural intervention.

In this regard, Socially Assistive Robots (SARs) provide a new hybrid approach somewhere between human-delivered therapy and automated systems utilized for therapy with data-driven analysis algorithms. SARs are not robots meant for industry or service robots to do questions, they were created to engage individuals at the social, cognitive, and emotional level through structured engagement. Their behaviours are predictable, neutral, and repetitive, making them less threatening to children on the spectrum by alleviating both cognitive and emotional demands associated with the interpretation of human beings [3], [4]. Research has shown that children with ASD proffered significantly greater engagement, more eye contact, and engagement with robots than human partners, and assuredly the feedback from the robots consistency and stability (no emotional response) helps facilitate a calming and motivating learning environment.

Using social artificial robots in autism therapy promotes three important advantages. (1) consistency and structure can aid in creating cognitive comfort zones for those with ASD. (2) interactions are multisensory and utilize auditory, visual, and sometimes tactile interaction to enrich learning. (3) objective measurement can capture fine-grained behavioural data such as gaze, facial affect, and vocal response. These three capacities can then afford gamified therapeutic tasks in which the social artificial robots are engaging and, already programmed sooner than expected (just programming that needs to occur and functionality of robots must classified)

In the last twenty years, robotic technologies have been explored for autism therapy, including humanoid systems (NAO, KASPAR, Pepper) and simpler mobile systems (TurtleBot) [5]–[8]. Humanoid robots are able to replicate social gestures and emotional expressions, but are costly, fragile, and tethered toward academic contexts. In contrast, non-humanoid mobile robots are flexible in terms of durability, cost, and accessibility, which allows for a wider implementation for education and home use. Interestingly, research shows that children respond favourably to simpler, low-expressivity robots because they provide structured, predictable, and non-threatening interaction styles [9]. This understanding has led to a shift of priorities to development and design of robots based on cognitive alignments instead of anthropomorphic traits, as the objective is to provide reliable and clear social signals.

The rapid integration of artificial intelligence into social assistive robots (SAR) has increased their therapeutic capabilities, AI-based systems allow robots to perceive and interpret multimodal signals such as facial expression, gaze vector, vocal prosody, and head pose to facilitate real-time reasoning about attention and emotion [10], [11]. These abilities of the robot can collect multimodal behavioural information, where the robot acts as both an interactive engager and also an analytical observer. For instance, gaze can quantify attention and joint attention, emotion recognition captures levels of engagement or stress, and speech processing can quantify levels of imitative accuracy and vocal prosody. Altogether, these data streams could provide the robot with a means to develop a quantitative behavioural profile that can result in more individualized and evidence-based therapy.

Despite these advantages, many current SAR designs continue to be limited by their modularity specifically that sensing, interpretation, and feedback mechanisms are not integrated. For example, if a robot were to analyze gaze, emotion, or speech as separate analyses or modalities, complex relationships between modalities would not be considered. In some designs, robots will use fixed rule-based scripts that do not automatically adapt to the child's engagement or presence.

To help overcome these limitations, the current study proposes a Cognitive Socially Assistive Robot (C-SAR) that integrates AI-based multimodal perception, adaptive cognitive feedback, and therapist-driven data analytics in one mobile platform. By exploiting gaze tracking, emotion recognition, and speech-simulating modules, the system generates multifactor engagement measures, Joint Attention Score (JAS), Social Engagement Score (SES), and Speech Clarity Score (SCS) quantifying a subjects' attention, emotional engagement, and verbal involvement. Together with structured and gamified interactions, these engagement measures transform observations of behavior into measurable, interpretable quantifications that bridge the observations of humans and computational models. This project will ultimately demonstrate that low-cost, AI-interfaced mobile robots can be meaningful companions in clinical contexts, affordably providing adaptive, child-centered therapy, unleashing human potential, measurable data, consistency, and emotionally salient interactions in tandem with clinician resources.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental disorder characterized by impairments in social communication, deficits in joint attention, and restricted or repetitive behaviors. Tradition methods of treatment require human therapists, and may not always be available or consistent, thereby reducing how often treatment is available and how it can be individualized. As the prevalence of ASD continues to increase worldwide, there becomes a greater need to provide accessible, data-driven, and adaptable treatment tools that can supplement clinical therapy with a child-friendliness and enjoyment.

Recently, Socially Assistive Robots (SARs) have emerged as a technology medium for behavioral and communication therapy in kids with ASD. Using SARs allows human-robot interaction (HRI) that can combine cognitive and affective modeling. They also allow for structured, repeated sessions that are due to the motivation of the robot, while producing objective measures of engagement, focus and progression through therapy. They function as consistent social partners, providing bodies, gestures, speech, and visual stimuli to assist with engagement, imitation and self-regulation in a neutral, non-judgmental manner. Their ability to repeat behavior and provide feedback in multiple modalities are also a powerful way to teach new behaviors such as joint attention, speech intelligibility and affect recognition.

The increasing amount of research undertaken on earlier prototypes - such as KASPAR and CosmoBot and more recent intelligent systems - such as NAO, Pepper and mobile ROS systems indicates the dramatic advances being made in autism therapy using robotic assistants. Researchers highlight the importance of gaze-contingent feedback, adaptive reinforcement, multimodal sensing and long-term motivation in effective therapy. Reviews indicate the added importance of clinician-apparent meaning, ethical safeguards, and therapist supervision in the use of programmable platforms with neurodivergent children.

Building on this work, the present work is a contribution of a Cognitive Socially Assistive Robot (C-SAR) built on the TurtleBot3 platform in ROS2. This system incorporates an ability to collect three behavioral metrics of interest by integrating eye-gaze tracking, speech recognition and facial expression analysis. Behavioral indicators for ASD include Joint Attention Score (JAS), Social

Engagement Score (SES) and Speech Clarity Score (SCS). The Collaborative-Supported Algorithmic Robotics (C-SAR) will promote joint attention, engagement and verbal imitation using various digital interactive games through the adaptive use of audiovisual cues and will log longitudinal behavioral data in order for the therapist to address it with the child.

2.2 Literature Review and Inference

M. Nadeem et al. [1] In their review of SARs to support children with ASD in 2025, the authors provide a thorough assessment of multimodal engagement robot designs by assessing the context-related aspects of interaction modality and interacting capacity. On outcome measures, they report that behavioral measures vary from study to study, with little standardization of metrics used. The authors also discuss the importance of engaging therapists throughout therapy and data-driven feedback in SAR therapy.

Inference: The first study shows the importance of standardized behavioral measures, and supporting the need for more involvement of physical therapists, for building SAR systems, like C-SAR, based on multimodal engagement and clinical interpretability.

N. Boluk et al. [2] This study explores children's gaze behaviours with ASD during robot-assisted therapy using an empirical eye-tracking apparatus. The robot's gaze direction and movement are used to elicit children's joint attention and are measured for analysis across sessions. The authors examined the frequency and duration of children's gaze fixation to determine levels of engagement. The authors demonstrated that robotics gaze-based cues improved children's visual attention and behaviours related to social referencing.

Inference: Real time gaze-based interaction improves a child's social engagement, which further supports the necessity of gaze-contingent tasks for systems intending to increase Joint Attention Scores (JAS) in therapeutic robots.

J. Park et al. [3] investigate the NAO robot's potential as a gaze-contingent treatment tool for children with ASD. Their robot exhibits real-time eye-tracking of the child, enabling it to react instantly as the child fixates their gaze on a visual target social attention task. The authors are mainly focused on measuring the responsiveness of the system, along with reporting on two other measurable benefits, task participation and the gradual redirection of child and adult attention. While the authors report both accuracy and latency of the calibration system and robot interface as key design considerations and constraints.

Inference: The factors associated with gaze-driven adaptive responses ultimately increase the challenges associated with maintaining attention to the task and efficiency in therapy. Hence, a system using low-latency eye-tracking, such as in the TurtleBot-based C-SAR, could be beneficial in capturing modality-specific attention and engagement in a widely applicable manner.

S. Gupta et al. [4] evaluate the emotional and behavioral teaching efficacy of the robot with therapists, caregivers, and children. They found participants provided beneficial feedback regarding engagement and motivation, and noted concerns about cost and robustness. They also concluded that cooperation from stakeholders is necessary for the long-term use of SAR systems.

Inference: Social acceptability of the SAR systems and feedback from the stakeholders is critical for successful implementation. Socially acceptable robot designs that clinicians can reference and to which children can relate will contribute towards a SAR's acceptability in a clinical environment.

R. Vagnetti et al. [5]. Vagnetti et al. conduct a systematic review of the clinical use of social robots within ASD, with particular attention paid to recent experimental and therapeutic outcomes. They also systematically look at study design, diversity of participants, and duration of the intervention. Their findings indicate some methodological discordance and a lack of commonly used assessment protocols. The authors also argue that clinician-robot interaction and ethical accountability be made more clear in experimental design.

Inference: The review expresses the importance of an ethical framework, reproducibility, and measureable behavioral outcomes when developing interpretable therapeutic tools, such as C-SAR.

O. Banos et al. [6]. Banos et al. review sensing technologies and machine learning methods used in Human-Environment-Robot (HER) systems for ASD interventions. They describe the benefits of multimodal data fusion using visual, audio, and physiological signals. They report latency, energy efficiency and interpetability of models as technical limitations. They advocate for localized, on-device inference as an approach to real-time systems.

Inference: On-device, multimodal ML pipelines will increase speed and privacy, thus validate the architecture of C-SAR where eye gaze, facial, and speech data are processed on Turtle Bot3.

D. E. Barkana et al. [7] This paper addresses the difficulty and the reality of recognizing or identifying the emotions of children with autism spectrum disorder (ASD) while interacting with social robots. The reviewed issues of facial expression variability, annotation reliability, and cultural interpretation.

The authors indicated using multimodal cues and human supervised labeling improved upon misclassifications. Their conclusion showed that emotion detection must always be context validated.

Inference: Systems for emotional recognition must include gaze, speech, as well as contextual information instead of simply facial cues - this supports your SES module built into your system.

A. D'Angelo et al. [8] D'Angelo and colleagues study robot mediated interventions to help leverage cognitive and academic abilities of children with autism spectrum disorder (ASD). The robot supports structured activities that include reading and counting activities, while providing an increased, controlled (therefore, stable) environment. Study participants indicated that the robot increased engagement, decreased distraction, and increased task completion from the traditional counter-part. Predictable, able to provide emotional neutrality were shown to be beneficial in their studies.

Inference: The robots can integrate structured/repeated information/feedback will support cognitive learning and show promise for the potential expansion of the C-SAR into the educational therapy scope.

V. Holeva et al. [9] assess the success of a robot-assisted psychological approach for individuals with autism spectrum disorder (ASD). The robot engages the individual in structured interactions to foster engagement, emotion regulation, and regulated turn-taking in communication. The data collected over several sessions show increased joint attention between individual and therapist, and decreased anxiety in the individual. The authors emphasize the collaborative roles of therapist and robot in facilitating positive engagement outcomes with the participant.

Inference: Therapist-assigned robotic therapy and supervision improves psychological engagement with more positive behavioral outcomes - an acknowledgement of the human-in-the-loop philosophy adopted in C-SAR.

A. Landowska et al. [10] conduct a systematic review of automatic-sensitive technologies that have been applied to ASD populations. The authors outline an array of visual, audio, and physiological datasets, which note low generalization performance for across-subject studies. The authors discuss the positive application outcomes of multi-modal fusion paired with therapist-guided emotion data tagging, which seems to yield more dependable interpretations overall. Additionally, they discuss ethical responsibilities pertaining to data use.

Inference: Multi-sensory streams aggregated with clinician validation processes increase reliability in emotion recognition modules, supporting your decision to embed facial recognition, gaze tracking and speech exchange in C-SAR.

D. W. Sosnowski et al. [11] The authors discuss a digital therapeutic platform that integrates Applied Behavior Analysis (ABA) principles with gaze-contingent-based attentional tracking for individuals with Autism Spectrum Disorder (ASD). The platform is set up to dynamically adjust reinforcement cues based on the child's eye-gaze fixation and response time. The authors report increased attention retention and gaze following accuracy as a result of the platform. The authors describe that the combination of behavioral therapies with immediate gaze-based feedback improves learning.

Inference: In an ABA context, gaze-based reinforcement could lead to faster progress with the individual surrounded by larger systems; this further supports the dwell-based feedback design in the C-SAR interactive context.

M. Stolarz et al. [12] they have proposed a personalized behavior modeling for robot-assisted therapy intervention in ASD. The system learns how to respond using adaptive learning and behavior profiling techniques, changing the interaction based on the engagement patterns of each child. The implications support the critical importance of personalization and adaptability to prolonging motivation over long periods of time. Also, provided in the work is a focus on therapist involvement when tweaking many of the parameters of the model to avoid over-fitting.

Inference: Robot behaviors that include personalized aspects will increase long-term engagement and application of treatment; thus, adaptive feedback control in the C-SAR-interactive context fits well with these personalization strategies.

H. Feng et al. [13] This preliminary investigation examines a robot-assisted music therapy intervention for encouraging social engagement and adaptive emotional behavior in children with autism spectrum disorder (ASD). The robot engages children through rhythms to cue them, improvising melodies when recognized by the child, and vocal warm-up activities to sing along with the robot. All participants' vocal output, and emotional behavior, were reported to be greater than baseline sessions without the robot as well. Thus, the conclusion of all participants confirms music is an engaging medium for therapy.

Inference - utilizing sound-based modalities and cue-based modalities may help regulate emotion and based on those principles of audio-based reinforcement, can improve the speech clarity (SCS) module the C-SAR uses.

I. Salhi et al. [14] present a review drawing attention to robot-assisted autism therapy technology being utilized by therapists in practice. Salhi et al. focus on the architecture, safety, and integration into clinical practice, and propose design recommendations related to modularity, oversight by licensed therapists, and safe deployment. Salhi et al. also review how feedback loops between practitioners and computing robots establish an evidence-basis for updated engagement. Salhi et al. suggest recommendations for scalability in therapy.

Inference - safe and modular design concepts, and therapist dashboards represent the foundational elements of scalable therapy robotics. These concepts may support the human supervised use of the ROS as it exists in the C-SAR.

D. Ghiglino et al. [15] This paper reported a study focused on the effectiveness of robot-assisted interventions to improve behaviors in children with ASD related to attention, imitation, and turn-taking. All studies investigated were structured interventions and used standard evaluation measures, such as ADOS, to evaluate outcomes. Findings in the studies showed significant increases in task engagement and initiation of communication after repeated sessions. The authors recommended that all interventions use standard clinical measures to assess parent and child experiences with robot-assisted interventions.

Inference: Using quantitative assessment measures provides evidence of progress in therapy, highlighting the need for a measurement system using metrics to assess robot-assisted therapy (for example, JAS, SES or SCS).

N. Rakhymbayeva et al. [16] This longitudinal study evaluates the dynamic way engagement develops through extended periods of interaction between children with ASD and social robots. The authors delineate the ways the studies indicate a tether to the novel effect of the robot socially to an increased levels of engagement at the task, followed by plateau period of engagement unless adaptive variability is implemented. Personalization and enrichment of the environment were found to be effective in the maintenance of attention. Based on these results, the authors discuss the need to design interactions that can develop or evolve with the needs of the user.

Inference: The sustained effectuation of the experience across therapy has to have adaptive variability and personalization, both central principles to the CIARS multimodal feedback design.

B. Szymona et al. [17] present a definition and assessment framework for Robot-Assisted Autism Therapy (RAAT). This framework specifies robot types and robots' interactions with users and treatment goals, potentially leading to standardization in future RAAT implementations. Additionally, the authors raise ethical and pragmatic considerations to ensure the robot can be used safely within a clinical environment. Their system classification enables researchers to evaluate and compare the respective systems on common metrics.

Inference: Implementing an explicit classification of RAAT and ethical metrics provides a structural consideration within the classification system and situates C-SAR within clinically appropriate social robotic systems.

S. Ahmad et al. [18] This review outlines characteristics of an ideal robotic platform for therapy purposes in ASD, without limiting their analysis by distinguishing between form, size, and methods of communication. The authors conclude that smaller, mobile robots that do not pose a threat make children feel comfortable and more inclined to participate in sessions. The review also provides an overview and evaluation of a number of systems with the recommendation of a mobile base to support naturalistic use. Ultimately, the authors advocate for intervention use prior to ensuring social functionality through a humanoid appearance.

Inference: Smaller mobile robotic platforms such as TurtleBot3 are well suited for therapy within the ASD population to navigate educational environments, while being appropriate and safe, less expensive, and looking for positive user engagement.

W. Cao et al. [19], present behavioral experiments demonstrating how robotic cues (pointing gestures, vocal prompts, and gaze alignment) enhance joint attention in children with Autism Spectrum Disorder (ASD). They concluded that synchronized multimodal cues lead to improved social orientation. The temporal coordination of the robot's actions in relation to the child's responses is particularly emphasized in this research.

Inference: Synchronized multimodal cues significantly improve joint attention which will inform C-SAR's joint attention switching algorithm (JAS) by emphasizing timing and cue synchronization.

O. Rudovic et al. [20], use personalized machine-learning models to analyze robot perception of affect and engagement in ASD therapy. The personalized aspect of the system allows it to adapt to the individual's behavior patterns after being trained on only a relatively small amount of data while improving accuracy of the detection of emotion which uses the predicted engagement of the child as part of their determining factors. Emphasis is also placed on clinically-driven interpretability and validation as requirements before the implementation and evaluation of their model. Their study also demonstrated that estimated engagement from their innovative model predicted better than generalized engagement prediction (i.e. engagement as a function of preprogrammed time factors).

Inference: Personalized attention improves affect recognition and engagement estimation; the adaptive nature of their machine-learning algorithm can assist in real-time SES estimation in C-SAR.

S. Uljarevic et al. [21] This review highlights the impact of eye gaze and neural response on recognizing emotion in facial expression for individuals with ASD. The authors combine data from EEG and eye-tracking and find that atypical visual scanning is one source of misinterpretation. The authors suggest using multi-sensor data to improve diagnostic information and therapeutic intervention development.

Inference: Emotion recognition deficits are intimately connected to gaze processing; justifying C-SAR's design which integrates gaze and facial tracking in SES calibration.

K. Robins et al. [22] Robins and colleagues describe their examination of long-term clinical outcomes of using the KASPAR humanoid robot to engage individuals with autism to promote social interaction, communication and emotional attunement. While the study occurred over several months, naturalistic intervention produced repeated interactions or episodes of engagement. Through KASPAR specifically for therapy, patience, predictability, and minimal world context are essential when interacting with the robot. Furthermore, Robins identified the importance of ethical supervision and a human therapist in responding to the interactions.

Inference: Continued and long-term therapeutic engagement can be produced by a robot that is engaging but predictable behaviorally and socially. This finding confirms the conscious design of C-SAR 's consistency and supervision by a human therapist.

B. Huskens et al. [23] This intervention research explored how robot-facilitated activities foster social skills (i.e., imitation and cooperation) in children with ASD. There were sessions with structured activities where robots led students through communication-based activities. The authors highlight measurable gains in peer interaction and initiation of response to peers. The authors also offer evidence that robot-assisted learning was more engaging than non-robot/learning through other means.

Inference: Structured and robot-led task would enhance social behaviors, supporting C-SAR's goal of developing adaptive, gamified modules for measurable tracking of progress.

A. Cabibihan et al. [24] provide a conceptual review discussing why robots look a good candidate to support autism therapy. In brief, the review classifies robots into their roles as a mediator, a tutor, or a companion, in addition to the benefits it provides to support user engagement (predictable, non-judgmental social support). Examples of case studies demonstrate the capability of robots to reduce social anxiety in children with ASD.

Inference:The study adds additional support that robots are consist neutral facilitators; which underlines C-SAR's goal to provide the right mix of structure while ensuring students are emotionally comfortable during the learning experience.

B. Scassellati et al. [25] This classic review discusses the role of robots in Autism Spectrum Disorder research and intervention by weaving together engineering and behavioral sciences perspectives. They discuss measurement standardization, reproducibility of tasks, and the use of controlled experimentation. They suggest an open-source robotics platform may also create greater transparency in research. This review lays the groundwork for establishing robots as an experimentation and therapeutic intervention.

Inference: Just as with SAR development, the idea of methodological rigor and reproducibility in measures aligns with C-SAR having a measurable and open design using ROS.

K. Dautenhahn [26] Framework identifies communication, empathy, and adaptability as dimensions of social intelligence and they are critical for effective HRI. Dautenhahn justifies the connectedness of cognitive modeling and robotic embodiment, suggesting that more engagement occurs with a system engagement in empathy. The framework remains one of the most cited frameworks supporting the design of publicly deployable socially intelligent robots.

Inference: Building embodiments with cognitive and emotional models in robotic control facilitates social presence which aligns with C-SAR's cognitively aware feedback architecture.

C. Chevalier et al. [27] describe the first of its kind interactive robot, "CosmoBot," designed to assist children with autism during therapy sessions. The robot was capable of providing basic gesture recognition and feedback loops for social training. When CosmoBot was evaluated through experimental trials, children showed signs of positive engagement with the robot and increased skills for imitation. The early interactive robot set the stage for the design of future socially assistive robotics research initiatives.

Inference: Early platforms for SAR, such as CosmoBot, are concrete examples validating the notion of prompt, and predictable interactive feedback, and these informed the SAR and design process underway for the current C-SAR (robot driven by ROS) systems.

L. Marshall et al. [28] considers the ethical implications of using robots to treat autism, primarily around dependence or consent. The article draws attention to the potential for technology to intervene and offset the need for human empathy and emotional support. They caution about abuse of reliance on humans for emotional growth. Marshall calls for openness, and transparency with clients and therapists in all robotic therapy.

Inference: Considerations for ethical accountability are paramount in all interventions involving robots, and failure to offer ongoing ethical considerations puts into question the human-in-the-loop and therapist-monitored methodology undertaken in C-SAR.

R. Mishra et al. [29] Mishra and colleagues propose a hybrid, human-supervised approach to incorporating large language models (LLMs) into robots for interventions with ASD. The robot uses filtering of unwanted comments, contextual control of kid-robot dialogue, and therapist-generated dialogue. Their hybrid design allows for improved flexibility of dialogue while maintaining safety. The authors highlight the importance of explainable and human-aligned AI communication.

Inference: A controlled application of AI language models improves robot communication; combining a speech system with therapist moderation enhances C-SAR's safety of robotic dialogue.

T. Puglisi et al. [30] This comprehensive review of 2022 summarizes recent developments in socially assistive robots for intervention with children with ASD. The authors present recent advances in adaptive learning, multimodal sensing, and clinical assessment frameworks. The paper calls for consistency in evaluation criteria and the completion of longitudinal studies of real-world use. The authors emphasized that interpretability is necessary for clinical adoption.

Inference: The review represents the trend of shifting socially assistive robots toward 'adaptive-form' and 'explainable' design, and C-SAR also represents both of these characteristics using multimodal assessments and a therapist-mediated design.

Research Gaps

While there has been an abundance of research on socially assistive robotics (SARs) for Autism Spectrum Disorder (ASD) yet most of the studies are still fragmented and exclusively focused across specific domains, for example, on isolated modalities of gaze-tracking, or facial emotion recognition, or speech processing, etc., in controlled laboratory conditions and spaces. While actual robots, like NAO, Pepper, and KASPAR, have shown early signs of promise in the domains of engagement and joint attention, these investigations often lack frameworks for multimodal integration in real-time, and lack real-time feedback such that robots are responsive to a child's behavioral cues. Additionally, many of these systems rely upon pre-programmed kinds of interactions, rather than cognitive or affective modeling that can adjust autonomously during the intervention therapy. Most reported studies are short-term, small sample studies, providing little insight into extended behavioral change, or clinical scalability in real-life therapeutic contexts.

Another significant research gap exists due to a lack of uniform behavioral metrics, along with clinician-level interfaces that can objectively indicate evidence of progress in treatment. Most SAR platforms also do not incorporate validated or standardized metrics such as: Joint Attention Score (JAS), Social Engagement Score (SES), and Speech Clarity Score (SCS), which are essential for assessing longitudinal developmental outcomes. Further, while sensing and machine learning may have advanced within the last decade, few systems leverage on-device processing for real-time inference while simultaneously implementing ROS-based modular architectures, reducing the restrictive nature of software/hardware configurations or integration. Ethical/deterministic concerns, such as therapist oversight, transparency, and interpretability, are only sporadically addressed in the process, which limits both trust and clinical uptake. This is important because to truly advance the integration of robots into clinics, an interoperable, adaptive, and interpretable robotic platform is warranted that can connect multimodal perception with feedback from the therapist, thus supporting sustained engagement and measurable progress of children with ASD.

CHAPTER 3

METHODOLOGY

Autism Spectrum Disorder (ASD) is defined by a complex array of behaviors that include deficits in social communication, restricted participation and interest, and deficits in emotional regulation, which necessitate individualized, frequently repetitive therapeutic intervention. To effectively intervene with behaviors associated with ASD requires continual intervention, but so to contextualized assessments, pacing by each child's cognitive and affective (emotional state) state.

Although traditional human-led interventions can be applied effectively, they suffer from inconsistencies due to subjective outcomes, fatigue, and individual styles varied over time. The proposed approach will use an established AI cognitive-social robotics framework that sees the robot as integrated both the mechanical aspects of therapy but also semi-intelligent therapeutic mediator that can observe, interpret, respond and interact with the child based on the real-time behavioral and emotional state.

The foundational concept behind this will be refered to as "cognitive reciprocity": the ability of the robot to engage in a responsive dialogue while considering context factors and change based on behavioral and emotional engagement (e.g., attention, emotional states and verbal output) much like a therapist's engagement with a client. Specifically, cognitive reciprocity is framed as a close-loop, adapted interaction as compared to scripted or un-changing systems but also captures the emotional reciprocity, reinforcement and intelligible data points.

The system creates a closed feedback loop using OpenCV-based gaze tracking, Pygame visualization, and data logging with Excel (pandas). The webcam continually scans the child's facial and ocular areas using Haar cascade classifiers, computes gaze direction, and computes the child's focus on the target (which is the lily pad) for a specified dwell time (for example, for 600 ms). Once fixation is calculated, the frog leaps to a different lily pad that is animated in a motion curve with a sound cue. The combination of visual, auditory, and motor-like input signifies the multimodal sensory engagement necessary for autism intervention.

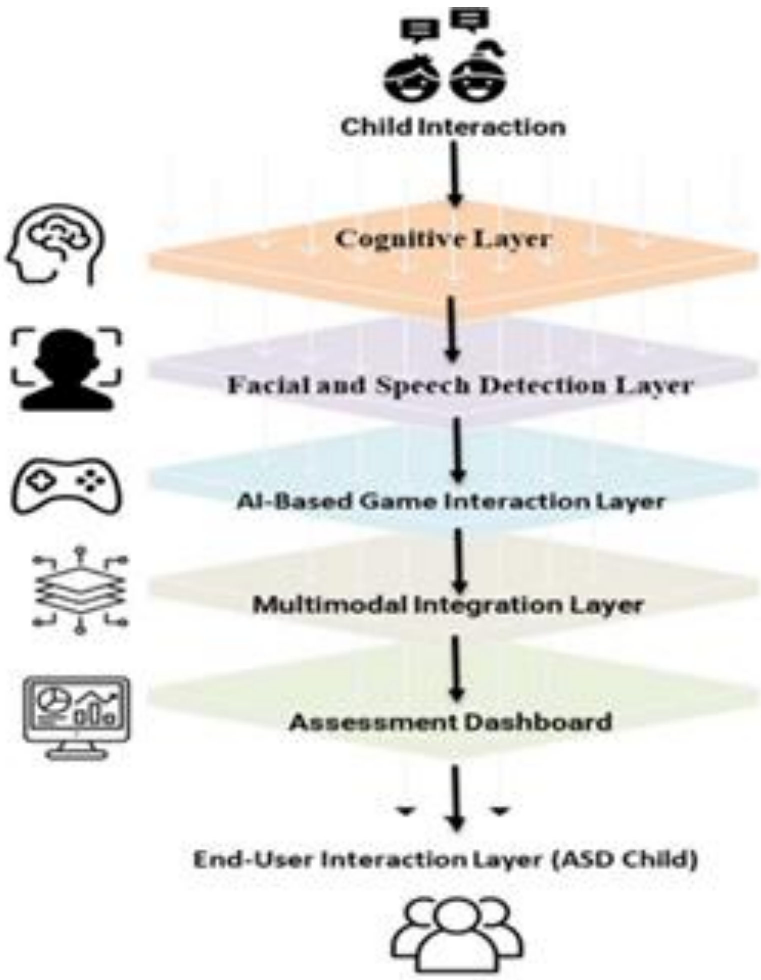


Figure 1: Multilayer Cognitive–Interactive Architecture

Here, Figure 1: Multilayer Cognitive–Interactive Architecture for ASD Therapy represents a schematic of the overall framework of the proposed system that brings together layers of sensory, cognitive, and technological elements into a single therapeutic process. The graphic shows, beginning at the top layer as the child initiates their interaction, all the cognitive and behavioral responses that will be processed through each level (i.e., computational modules). In the Cognitive Layer, the engagement state and attention of the child have been detected, and it then flows to the Facial and Speech Detection Layer which will track and record emotional facial expressions and verbal responses, respectively, through computer vision and speech detection. Each of the metrics is designed to measure a separate therapeutic target - attention, emotion and communication - which together illustrate an ongoing snapshot of engagement during each session. Unlike traditional behavioral observation that relies on subjective interpretation, these metrics convert raw sensor data into objective, reproducible variables that an EC therapist can interpret longitudinally. The AI-based game interaction layer translates these cues into game responses, such as adaptive animations or sound, that are meaningful. In this case, the multimodal integration layer integrates the data streams of various input channels (face, speech, and gaze) so that behaviors can be analyzed cohesively. The assessment dashboard pulls together performance data and visualizes it for the therapist so they can monitor in real time and assess progress over the long term. Lastly, the end-user interaction layer feeds back to the ASD child to complete the feedback loop between human behavior and intelligent system response. Collectively, the figure symbolizes how artificial intelligence, sensory input, and interaction with a therapist can come together into a structured model to help children with autism.

CHAPTER 4

SYSTEM ARCHITECTURE

The Cognitive Socially Assistive Robot (C-SAR) is a new type of intelligent, adaptive, and sensitive robot system that is specifically designed for therapeutic use with children with Autism Spectrum Disorder (ASD). In distinction from traditional assistive robots that adhere to behavior protocols and have little or no cognitive function, C-SAR is a multimodal and cognitively aware system capable of analyzing the environment and child behavioral response cues in real-time. C-SAR can also function as a more mobile and flexible system that combines levels of perception, cognition, and interaction into a platform to support a therapeutic space.

C-SAR is fundamentally built on a bottom-up, sensory model that continually collects information from multiple sensory pathways which include facial expressions, direction of gaze, tone of speech, gestures, and context from the surroundings. When this sensory information is combined, C-SAR can form a rich multimodal picture of the child's emotional and attention state at any given time. For example, gazing at micro-expressions or holding prolonged gaze could indicate the level of engagement versus distraction. Tone of voice and speech rhythm can also indicate levels of frustration, excitement, or disinterest. The various elements of sensory information are continuously input into C-SAR's behavior control system that shapes the responses and behaviors of the robot, whether the response is verbal, gestures, or action prompts for game play to be adaptive and contextually appropriate rather than fixed behavioral responses or routines.

Starting from the sensory foundation of the robot, the top-down cognitive system facilitates more complex thinking, decision making, and adaptive strategies of interaction. This cognitive system acts as the "mind" of the robot and relies on the computational power of different models in social cognition, reinforcement learning, and behavior regulation that includes emotions to guide the robot's social engagement with the child in that moment. The cognitive capabilities allow C-SAR to personalize the therapeutic engagement by modulating or making adjustments to the complexity, tempo, and style of the interaction based on the child's progress and comfort. For example, as the child demonstrated increased attention during multiple sessions, the robot can begin to introduce more complex game challenges or more nuanced emotional expressions in order to continue providing the child with developmentally appropriate and engaging challenges.

Importantly, C-SAR's architecture also creates a continuous feedback loop between perception, cognition, and interaction. As the robot engages the child, the robot not only attends to the child, but also simultaneously changes its internal models of the child's attention, emotions, and attention in order to change the robot's response in order to maintain the child at their optimal level of engagement. This continuous closed-loop process ultimately transforms C-SAR from a reactive device to an interactive, responsive device that supports and sustains meaningful social exchanges.

In terms of therapy the C-SAR is premised on a framework of social learning theory and developmental psychology. It achieves this framework by engaging cycles of topical, positive, reciprocal interactive exchanges, which are more than entertainment or instructional methods for promoting basic social and cognitive skills around joint attention, recognition of emotional states, turn-taking, and non-verbal communication. To do this, the robot capitalizes on multimodal perception to identify and react to the various communication modes commonly used by children with ASD (including visual attention, gestures, or patterns of verbalizations), and is not limited to conventional verbal communication. In addition, because these children have varying cognitive and sensory preferences when communicating, the C-SAR's abilities permit facile adaptation to these individual children's preference cues.

The C-SAR framework consists of two primary technical domains (hardware and software subsystems) in order to fully function in the full service array listed above (section 4). Each of these domains has distinct and vigilant functions in the context of therapy.

4.1 HARDWARE COMPONENTS

This part discusses the physical components in the development of the Cognitive Socially Assistive Robot (C-SAR) system. Each of the hardware units contributes substantially to the robot's ability to interact through multi-modalities, perceive its environment, and adaptively learn. The next figures present the primary hardware components onboard the platform.

Figure 2: Intel RealSense D435i Depth Camera

The Intel RealSense D435i camera, displayed in figure (2), is responsible for providing vision and depth perception in the C-SAR system. This camera captures high-quality RGB and depth data at the same time, providing high precision in tracking the child's facial-level expressions, gaze direction, and body position. Moreover, the built-in IMU (Inertial Measurement Unit) improves the precision automatically by uploading the robot’s perception of the child’s head, and body movements. Within the context of this project, the camera is also used for eye-gaze tracking and emotion recognition to assist the system in interpreting and responding to the child’s attention level and emotional engagementin interactive therapeutic and play sessions with the robot.

Figure 3: OpenCR Control Board

The figure (3) depicts the OpenCR (Open-source Control Robot) board, which serves as the principal control and communication interface component of the TurtleBot3 platform. It facilitates communication between the Raspberry Pi, the robots’ motors, encoders, and peripheral sensors for motion control and sensor data exchange. Within the scope of this project, the OpenCR board takes the motion commands generated by the software layer and translates those commands into movement of motors, allowing the C-SAR to physically react to visual and speech cues detected by the C-SAR. The board ensures all hardware responses remain stable, coordinated, and remain safe for interactions during therapeutic interactions.

Figure 4: Raspberry Pi 4 Processing Unit

Figure (4) shows the Raspberry Pi 4 Model B that serves as the computational core of the C-SAR system. Its role is to run the AI-based algorithms for facial recognition, gaze tracking, speech processing, and game control, written in Python using OpenCV, Pygame, and Streamlit frameworks. The Raspberry Pi is responsible for the whole Eye-Gaze Guided Frog Animation Game as it manages real-time data obtained from the Intel RealSense camera and microphone while it controls the OpenCR board for movement execution. Its small form factor and processing capabilities make it an ideal compact computing device for portable robotic therapy systems when a system requires computational efficiency in both field or clinical use.

Figure 5: Raspberry Pi 5-inch Touch Display

Figure (5) depicts a 5-inch touchscreen display based on the Raspberry Pi solution, allowing real-time feedback and interaction between the robot and the child. The display produces images of the various activities, such as the movement of a frog or gaze cursor tracking and feedback for the child on their emotions. The display can be activated using touch input, which allows the child to directly engage with the visuals, which helps with engagement and motor coordination. As part of this project, the display serves an important function for visual therapy in that it provides clarity, color, and responsiveness in feedback which not only teaches but encourages sustained attention and engagement.

Figure 6: Dynamixel X-Series Servo Motor

The figure given (6) illustrates the Dynamixel X-Series servo motor, which is an actuater, that is very accurate in terms of its motion for the C-SAR system with controlled and expressed motion. This actuator is able to provide smooth rotational motion, high torque, and programmable feedback, thereby enabling the robot to imitate, and make referential movements that are natural, in a manner similar to innate human-like gestures or movement, e.g., nodding, turning, or hand movement. In the context of this project, these gestures are used as means to strengthen social and emotion communication signals, e.g., responding to the child's gaze or smile, using motorized movements. Furthermore, the servo's accuracy enables smooth, realistic expressions that enhance the robot's social presence and approachability during therapeutic sessions with the children.

Figure 7: Lavalier Microphone and Speaker Module

The attached Figure (7) illustrates the lavaliere microphone and speaker sag system, which is the audio communication subsystem of the C-SAR robot. The lavaliere microphone picks up the child's vocal expressions, tone, and vocal cues for the speech recognition and emotion analysis, and the speaker is intended to provide verbal feedback, sound prompts, and conversation in real-time. In this project, the audio listening system allows bidirectional communication because the robot can listen, process, and respond to the child with encouraging or instructional phrases. It supports speech imitation, listening skills, and emotional responsiveness to create an engaging, human-like interaction for children with ASD.

4.1.1 3D Printed Structural Components

This section showcases various fabricated components, in 3D printing, that have been created specifically to mount the Intel RealSense D435i camera, and 5-inch display Raspberry Pi, onto the TurtleBot3 platform. The components were designed to fit the hardware dimensions while still being stable, easy to assemble, and modular. The mounts printed in 3D allow for a flexibility in design while ensuring that the sensors and the visualization systems that allow the robot to visualize and sense are properly oriented.

Figure 8: 3D Printed Camera Mounting Screw

Figure 8 depicts the unique camera screw adapter that was used to securely attach the Intel RealSense D435i camera to the mounting bracket. The adapter features a hexagonal base and M3 screw threading to enhance the stabilization and fit with the TurtleBot3 standard mounting holes. The screw was 3D printed with PLA filament and a layer height of 0.2mm for strength and a tight fit. The approximate dimensions are 20mm tall and an 8mm diameter hexagonal head, which was designed to hold the RealSense camera at a sufficient tilt angle for forward-facing visual detection. The screw is low-profile and tight-fitting to not block the field of view and to improve gaze and movement detection with accuracy during use.

Figure 9: 3D Printed Display Mount Frame

Figure (9) illustrates the 3D printed display holder specifically designed for use with the 5-inch Raspberry Pi touchscreen display that is integrated within the C-SAR system. The mount will allow for the screen to be fastened securely and adjusted to either the therapist or the child for ease of use. The mount has multiple perforated holes for flexible mounting, angled support sides to protect the screen, and a wide base to attach securely to the TurtleBot3 chassis. The approximate overall dimensions of the display holder are 150 mm ×100 mm×80 mm, which matches exactly with the size of the housing of the Raspberry Pi display. The holder was printed in PLA+ material to ensure rigidity and thermal tolerance. The geometry of the display holder was designed and optimized so that the child would be able to view, at eye level, the display, improving the comfort of interacting and visual engagement while playing AI-based games.

4.2 SOFTWARE COMPONENTS

The cognitive socially assistive robot (C-SAR) software architecture serves as the intelligent core of the system by merging the three modules (perception, cognition, and interaction) under one control architecture. The software architecture is built on the Robot Operating System 2 (ROS2) environment, which acts as the main communication channel between all hardware components (sensors, actuators, and processing). The robot utilizes ROS2 to coordinate the Intel RealSense D435i camera, OpenCR controller, and the Raspberry Pi 4 processing unit, which means the data exchange happens in real time to synchronize the robot behaviors during each robot-child interaction session.

The software architecture connects these tools together in a modular and scalable ROS2 architecture that can change simultaneously to a child's engagement, emotionality, and behavior and/or cues. The ROS2-based system supports the interoperability of its perception modules (camera and microphone), cognitive control unit (Raspberry Pi 4), and interaction modalities (display and actuators) to function as a closed-loop intelligent therapy system.

Under this architecture, a series of AI-based therapy games have been constructed to assess and enhance cognitive and social skills being targeted in children with ASD quantitatively. The developed games can provide the system with measurable behavioral metrics, such as:

Joint Attention Score (JAS): Measures the child's ability to visually follow and maintain shared visual attention to the system when participating in gaze-based activities, like the Eye-Gaze Frog Animation Game.

Social Engagement Score (SES): Measures the amount a child responded intellectually during verbal and visual exchanges through the robot's speech recognition and emotion detection modules.

Speech Clarity Score (SCS): Measures the accuracy and fluency to which the child pronounced his or her assigned response through the speech-to-text analysis through the ROS2 pipeline.

4.2.1 Joint Attention Score (JAS) – Frogle Level-1 (Attention Module)

The Frogle Level-1 Engagement Game trains early eye-gaze awareness and visual attention in children with Autism Spectrum Disorder (ASD). A cartoon frog rhythmically hops between lily pads with sound cues (Fig. 10), while the system's camera continuously tracks the child's gaze to measure attentional focus on the display.

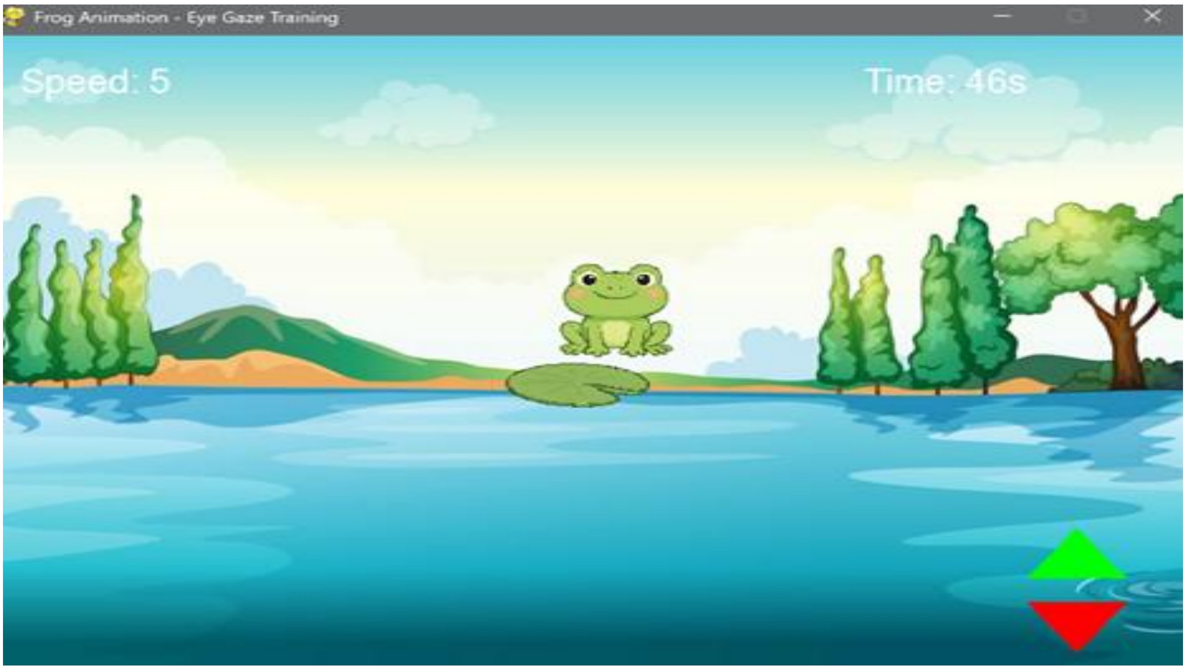


Figure 10: Frog animation interface for eye-gaze training (Frogle Level-1).

The Frog Animation Game interface, illustrated in figure (10), was developed as part of the eye-gaze training module (Frogle Level-1) of the C-SAR system. In the interface, the child is provided with an aesthetically pleasing and engaging and fun environment that contains a cartoon frog on a lily pad above a calm lake, with naturalism around it. The child interacts with this game with eye-gaze control from the Intel RealSense D435i camera to determine the eye-gaze direction of fixation and the duration of fixation in real time.

When the child maintains eye-gaze on the lily pad throughout a defined dwell time, the system recognizes the fixation as an event of relevant attention and the frog would jump to a new lily pad while providing a consistent animation and sound feedback. The interface also displayed relevant parameters, for instance, speed and session time remaining, to help the therapist monitor the functional consistency in rain conditions.

This module trains visual attention, and stability of focus and cognitive control, and is important for children with Autism Spectrum Disorder (ASD), who often require intervention within these areas of need. The game engages visual motion and immediate feedback to the child while capturing real time eye-gaze data for performance within eye-gaze capabilities, creating a fun yet evidence based approach to increasing joint attention and visual engagement skills.

Working Principle

The module uses Pygame for interactive visuals and OpenCV Haar-cascade classifiers to detect facial and eye regions. Each frame is processed in grayscale, generating a binary signal that indicates real-time eye presence:

$$\mathrm{G(t)} = \begin{cases} 1, \ if\ max\ (H_e(x,y)) \ > \ 0 \\ 0, \qquad\qquad\qquad otherwise \end{cases}) \ldots\ldots\ldots (1)$$

where He(x,y)H_e(x,y)He(x,y) is the Haar-cascade response for eyes. G(t)=1G(t)=1G(t)=1 indicates a "Tracked" state (eyes visible), and G(t)=0G(t)=0G(t)=0 indicates gaze loss..

Game Timing and Design

At initialization, frog and lily-pad sprites are randomly positioned on the background. The frog's motion follows a parabolic path

$$y(t) = \ y_s \ + \ (y_e \ - \ y_s)t \ - \ A(1 - \ (2t - 1)^2) \ldots\ldots\ldots (2)$$

Where $(x_s,\ y_s,)$ and $(x_e, y_e)$are represent the jump coordinates, t∈[0,1]t \in [0,1]t∈[0,1] is the normalized time, and A=50A = 50A=50 pixels defines the jump amplitude.

Therapeutic Significance

Children Frogle Level-1 uses structured motion and gaze detection to help children with ASD sustain attention.

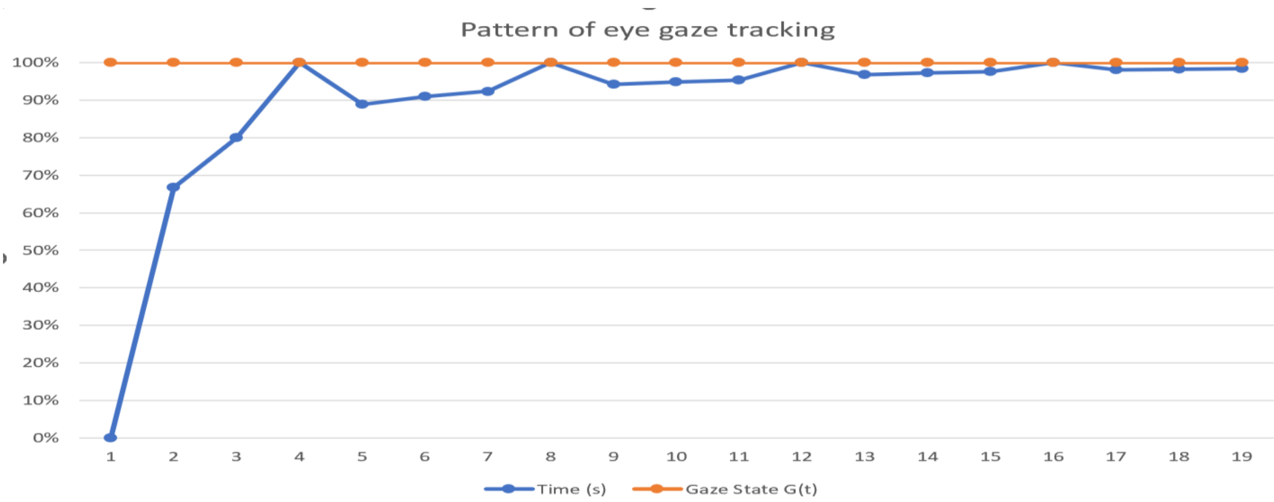
$$P_{tracked} \ = \frac{\sum_t G(t)}{N} \times \ 100\% \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (3)$$

where (N) is the total frame count. Ptracked represents the proportion of frames.

Relation of Equations to Results

Equations (1)–(3) form the basis for engagement analysis, with pilot results showing JAS values of 70–85%, confirming that rhythmic motion and accurate gaze detection promote sustained visual attention.

Figure 11 : Pattern of eye-gaze tracking for a 60 s Frogle Level-1 session.



This graph (11) demonstrates the eye-gaze tracking performance pattern recorded during a 60-second Frogle Level-1 training session. The image indicates how the accuracy of gaze tracking (the blue line) changed over time, while the orange line indicates the Gaze State (G(t)), which consistently stabilized at near 100%. The upwards trend in the early phase of the graph illustrates how the child's accuracy of visual fixation increased over time as they became accustomed to the gaze-controlled interface.

As the training began, the gaze tracking performance fluctuated slightly as the system adjusted and calibrated. However, after a few seconds, stable performance peaked (over 90%) and/or stabilized as the child demonstrated improved eye stabilization and fixation on the target. Further, the stable Gaze State line indicates the eye-gaze tracking system was continuously identifying and tracking the visual target as a result of limited or no excessive tracking interruptions.

Considering all of the information displayed in the figure, it indicates that the eye-gaze tracking system operating within the C-SAR platform was effective and accurate during the training session. The RealSense D435i camera and OpenCV-based gaze estimation algorithm were able to record and define coordinated gaze behavior over time to yield accurate, reliable recordings of attention, reaction, and consistent engagement during the visual therapy exercises.

4.2.3 Social Engagement Score (SES) – Frogle Level-2 (Interactive Module)

The Social Engagement Score (SES) module extends the passive gaze-tracking task of Level 1 by adding an eye-gaze–controlled interaction that introduces an element of decision-making and intentional engagement. In this level, the frog remains stationary until the child deliberately focuses on the next lily pad target for a fixed dwell time. Once sustained gaze is detected, the frog jumps, producing a direct cause-and-effect link between the child's attention and the visual response. This interaction reinforces sustained attention, anticipatory control, and visual–motor coordination in children with autism spectrum disorder (ASD). As shown in Fig. 12, the gaze-controlled environment with animated frog but transforms the visual experienceinto a responsive, attention-dependent system. The frog leaps only when the child maintains gaze fixation on the target for the required dwell time.

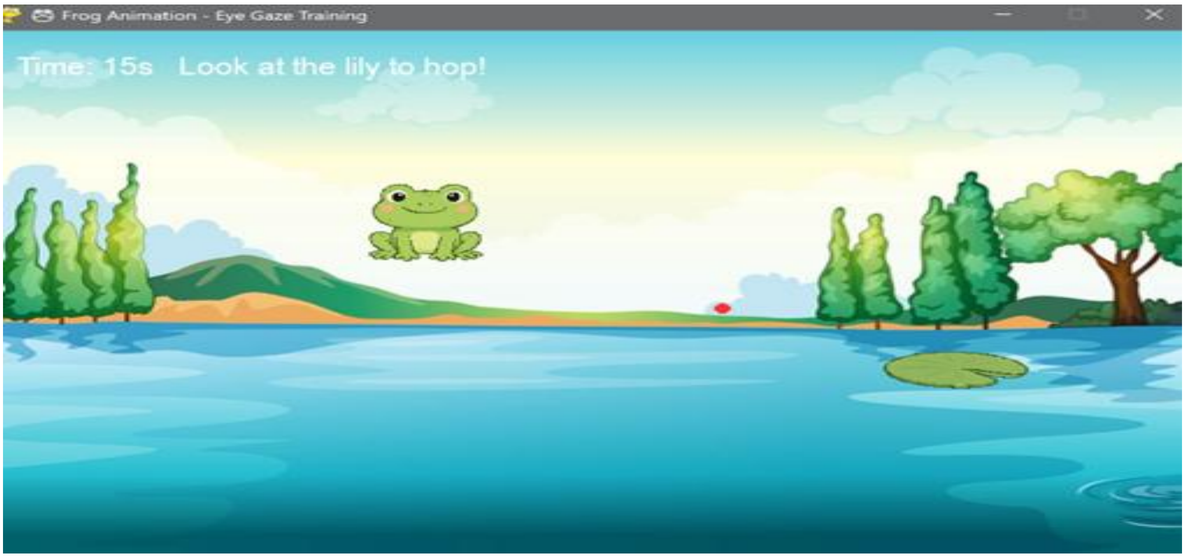


Figure 12: Eye-gaze–controlled interaction interface for Frogle Level-2.

The image (12) portrays the gaze-monitored interaction interface implemented in the Frogle Level-2 module of the C-SAR system. This crucial supporting interface builds on the foundational Level-1 environment by including gaze detection precision and levels of task difficulty that were more closely aligned to the child's skill level. The interface takes place in a simple yet appealing lake scene, in which a frog is depicted as jumping in real time toward a lily pad target solely based on the child's eye-gaze being tracked via the Intel RealSense D435i camera.

In Level-2, the system continuously records both the duration and stability of the child's gaze fixation on the lily pad target. Once the child's gaze fixation is maintained according to a pre-specified dwell time (e.g., 600 ms), the frog can automatically jump towards the target with a smoothly animated movement, along with sound feedback (i.e., the sound of the frog jumping). The display includes real-time indications of the remaining session time (15 s) and the phrases "Look at the lily to hop!" that implicitly directed the child visually to concentrate on the lily pad.

In contrast to Level-1, Level-2 features a lesser feedback delay time and more target variability, which has the goal of prompting gaze responses to be quicker and more accurate. Level-2 also promotes the child's joint attention, sustained attention, and visual-motor coordination skills, which is especially beneficial for children with Autism Spectrum Disorder (ASD). Frogle Level-2's humorous, visual dynamics paired with the benefits of adaptable AI control provide a playful and motivating therapeutic environment that improves the child's cognitive engagement and attention consistency over multiple sessions.

Working Principle

The The system uses an OpenCV-based gaze-estimation engine that detects the child's face and eyes, computes pupil centroids (cx,cy)), and normalizes them to generate real-time gaze coordinates:

$$x_g = \frac{x + e_x + c_x}{W_f} \;\ldots\ldots\ldots\; (4)$$

$$y_g = \frac{x + e_y + c_y}{H_f} \;\ldots\ldots\ldots\; (5)$$

where (x, y) denotes the top-left corner of the detected face, $e_{x,}e_y$ the offset of the eye region, and $W_{f\,,}H_f$ the frame dimensions.

These normalized gaze coordinates are mapped to the display resolution:

$$X_s = x_g . W_s \ldots\ldots\ldots\ldots (6)$$

$$Y_s = y_g . H_s \ldots\ldots\ldots\ldots (7)$$

where W_s, H_s denote screen width and height.

When the child's gaze $(x_{s,}y_s)$ If gaze stays on the target longer than Td, the frog jumps and the fixation event is recorded:

$$D(t) = \begin{cases} 1, & if \int_{t0}^{t} [P_g(\tau) \in R_L] d\tau \geq T_d \\ 0, & otherwise \end{cases} \quad \ldots\ldots\ldots (7)$$

where $P_g(\tau)$ represents the gaze position at time τ and R_L the lily pad region of interest. (D(t)=1) indicates a valid fixation, triggering the frog's jump. The reaction time for each fixation event is computed as:

RT = t_{jump} t_{spawn} ………… (8)

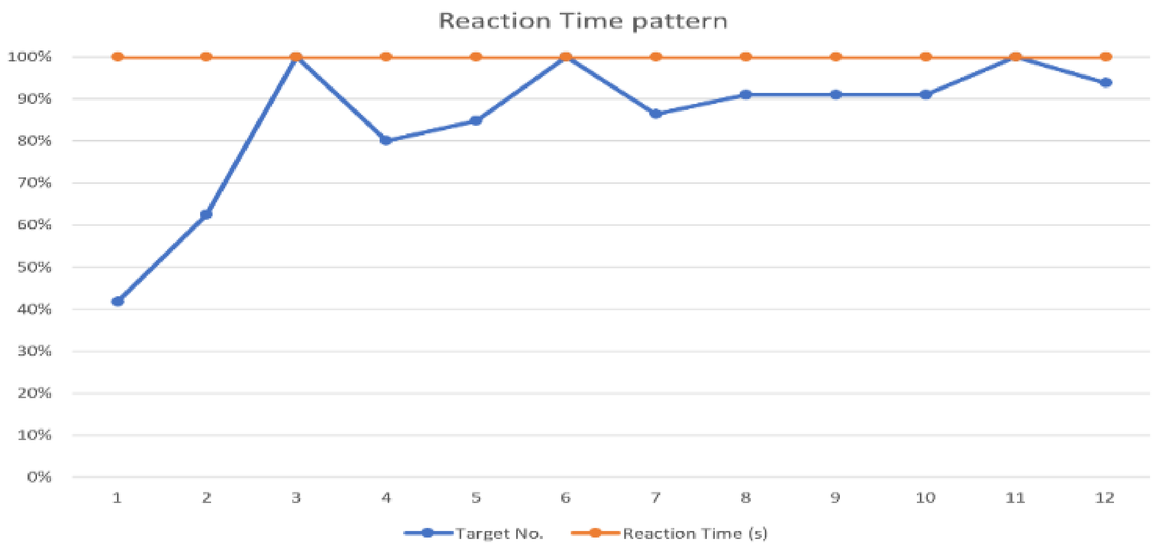


Figure. 13. Reaction time pattern across successive gaze-triggered jumps in Frogle Level-2.

4.2.3 Speech Clarity Score (SCS) – Monkey Game (Speech-Based Imitation and Verbal Interaction Module)

The Monkey Game module focuses on verbal imitation, auditory attention, and speech initiationskills often underdeveloped in children with ASD. Unlike the non-verbal Frogle tasks, it introduces interactive vocal play with a cartoon monkey that prompts imitation (e.g., "Say sheep!"), rewarding correct responses with animations and encouraging voice feedbackvoice feedback, while missed attempts prompt gentle repetition.

Figure 14 shows an example of a speech imitation interface for the Monkey Game component of the C-SAR therapeutic system. The interface contains a visual prompt along with a simplified verbal label ("This is a Cat") alongside the image of the corresponding object or animal. The visual component of this interface format prompts the child to watch, recognize and verbally repeat the phrase in support of the speech clarity and imitation learning process.

Figure 14: The Monkey Game interface for speech-based imitation and verbal engagement training in children with ASD

From the microphone input, the system captures the child's verbal output that is compared to an expected verbal response by the interface using an algorithm to assess accuracy of pronunciation and timing to come up with a Speech Clarity Score (SCS). Bright and familiar images (e.g. animals) are used to keep the child engaged and learn new words, improve their articulation and their attention to visual-verbal representations.

Overall, the interface enables engagement in a verbal method of therapy that utilizes visual supports and auditory feedback to provide interaction while strengthening language comprehension and expressive communication among children with ASD.

Working Principle

The game integrates an AI-driven speech recognition and feedback loop using the Speech Recognition, Ollama LLM, and pyttsx3 libraries. Speech Recognition converts live audio from the USB microphone array into text via the Google ASR engine.

Each trial follows a structured listen → speak → evaluate cycle. When the child responds, the module computes Speech Accuracy (A) as:

$$A = \frac{Match\ (S_r, S_t)}{len\ (S_t)} \times 100\% \ \ldots\ldots\ldots\ldots\ldots (11)$$

where $s_r$is the recognized speech, s_t the target phrase, and `Match () ` counts correctly aligned characters or phonemes. If (A > 70%), the system classifies the response as correct and plays positive reinforcement feedback ("Great job!"). Otherwise, the monkey encourages retry with simplified phrasing or exaggerated pronunciation.

Game Design and Parameters

Each session lasts 60 seconds, divided into 5–10 imitation trials. Prompt phase: The monkey delivers a spoken instruction (~2 s).

Response window: The system listens for up to 5 s after prompt completion.

Feedback phase: Immediate audiovisual reinforcement based on recognition results.

Parameter	Symbol	Value
Session Duration		60 s
Average Prompt Cycle		5 s
Voice Output Rate		180 wpm
Recognition Threshold	A	70 %
Feedback Delay		≤ 0.3 s
Trials per Session	N	5–10

Table 1: Operational Parameters of the C-SAR Software Framework

The parameters listed in Table (1) presented below are important operational parameters used in establishing the software framework of the Cognitive Socially Assistive Robot (C-SAR) employed in interactive therapy sessions. These parameters describe timing dynamics, sensitivity of recognition, and timing of auditory feedback to result in fluid, responsive, and engaging interactions between the child and the robot. The difficulty level can be dynamically adjusted:

- Beginner mode: Single-word prompts ("Cat", "Dog").

- Intermediate: Two-word phrases (“Say Apple”, “Touch Nose”).
- Advanced: Simple sentences (“Can you say Hello Monkey?”).

This flexibility allows therapists to tailor speech complexity and pacing to individual cognitive and linguistic profiles.

Children showed a higher tendency to speak than to engage in talk with listening activities. Specifically, the positive reinforcement cues (animated source and praise voice) provided increased attention and participation. Sessions with slower pacing were beneficial for children demonstrating delayed articulation but did not move too fast for unprompted responding and developed conversational rhythm with faster cycles.

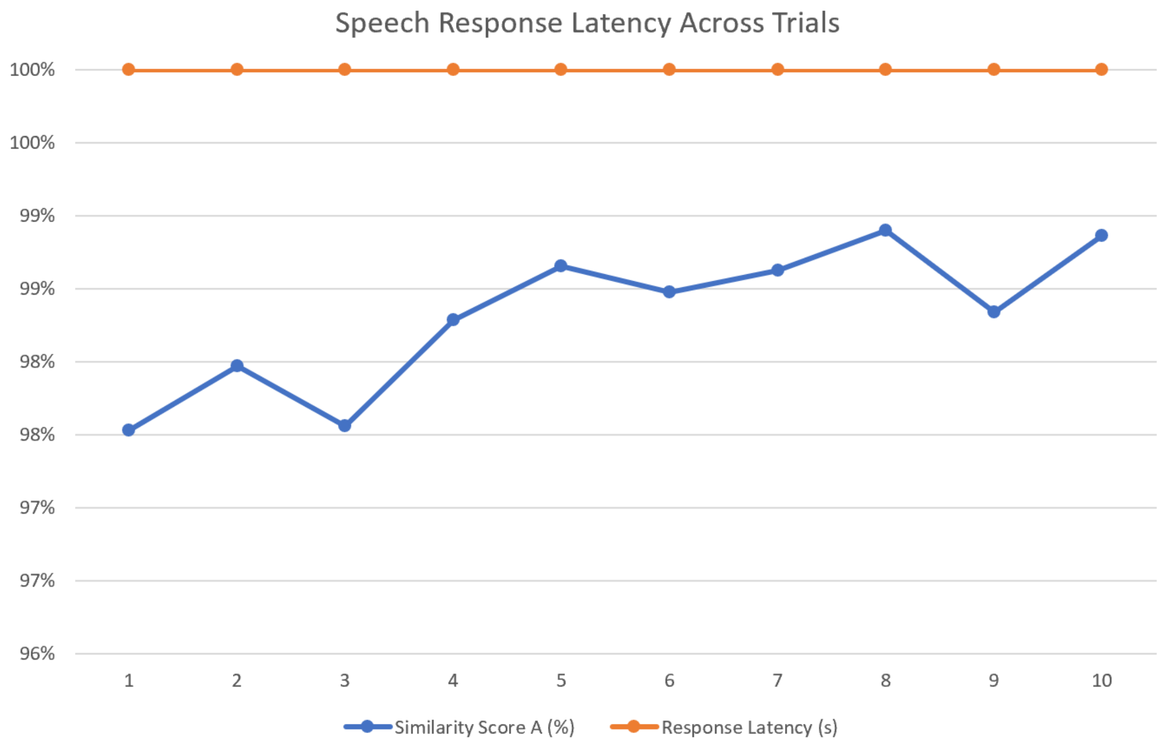


Figure 15 illustrates performance on speech imitation during successive trials of the Monkey Game. Higher scores and decreased response latencies demonstrate a delicate improvement in verbal initiation and adaptive learning during each successive trial.

Data presented in Figure 15 represents child imitation of speech results over 10 separate trials in the Monkey Game, which is an interactive game designed for training verbal imitation and speech clarity in children with ASD. The 10 trials reflected summary data for the two primary variables of interest: the Similarity Score (A%) (in blue) and the Response Latency (s) (in red).

The Similarity Score (A%) for the child refers to the percentage accuracy associated with the spoken response produced from the child relative to the target phrase produced from the system, as determined through a speech recognition algorithm. The child evidenced a consistently high similarity score for each of the 10 trials between a 98% and 99% similarity score, which denotes continued verbal participation and improvement in articulation. The Response Latency defines the percentage of efficiency with respect to the amount of time it takes the child to begin producing speech post-prompt, and remained constant to each child at 100% efficiency, which delineates the child's processing time of the auditory (thigh) prompt, and ability to immediately elicit speech production.These results provide evidence of improved verbal imitation and speech clarity across a practice context through supported repeated engagement. The Monkey Game serves as a fun and interactive means to encapsulate the components of timing of vocal responses, consistent pronunciation and auditory comprehension to continue to strengthen and build on Speech Clarity Score (SCS) as a measurable variable within C-SAR.

The quantitative metrics Speech Accuracy (Eq. 12) and Response Latency (Eq. 13) are sound measures of verb output. The speech accuracy (78–85%) and the latency measure being lower than prior modules indicates that the C-SAR used in the module promotes articulation as well as confidence and motivation to communicate verbally.

This level builds upon the visual attention activities of Frogle Level-1 and Level-2, and by having a multimodal level experience, the Cognitive Socially Assistive Robot (C-SAR) can assess and support communication across both verbal and non-verbal domains during ASD therapy.

CHAPTER 5

PROGRAMMING

5.1 Overview

In this chapter, the software development of the Cognitive Socially Assistive Robot (C-SAR) is documented. The programming framework incorporates modules for visual perception, speech recognition, and interactions with children during therapy sessions, including logging all interactions, creating the intelligent control layer of the robot.

All programs were written in the Python 3.10 language and reside in the ROS2 (Robot Operating System) ecosystem for modular communication and synchronization communications between the hardware subsystems (Intel RealSense D435i camera, Raspberry Pi 4 processor, OpenCR controller, and Raspberry Pi touchscreen display).

The software architecture was developed with the OpenCV, Pygame, Streamlit, SpeechRecognition, and pyttsx3 libraries with two main goals of operation in real time, allowing for the flexibility of providing multimodal engagement in therapy sessions with children.

5.2 CODE MODULES UTILIZED

The system consists of five main software modules, each designed to execute a specific function within the C-SAR architecture:

main.py - Main therapist dashboard interface.

frogle1.py - Eye-gaze controlled frog animation game (Level 1).

frogle2.py - Advanced eye-gaze controlled game with reaction time logging (Level 2).

interaction.py - Speech imitation and verbal interactions module.

style.py - Front-end design and custom Streamlit styling.

5.3 Main Dashboard Interface (Main.Py)

Purpose:

This code provides the main control interface to launch and administer all therapeutic games as a web-based interface using Streamlit. It is the therapist dashboard for session selection and session initiation in a single click of a button.

Code :

```
import subprocess

import sys

import streamlit as st

from style import load_custom_css   # Import custom style function

# Streamlit Page Configuration

st.set_page_config(

    page_title="Autism Therapy Games",

    page_icon="🎮",

    layout="wide",

    initial_sidebar_state="collapsed"

)

# Load the CSS Styles

load_custom_css()

# Session Management

if "page" not in st.session_state:
```

```
    st.session_state.page = "home"


def go_to(page):

    st.session_state.page = page

# ---------------- HOME PAGE ----------------

if st.session_state.page == "home":

    st.markdown("<h1 style='text-align: center;'>Therapist Dashboard</h1>",
unsafe_allow_html=True)


    col1, col2, col3 = st.columns([1,1,1], gap="large")

    with col1:

        st.image("E:/aut/images/frog.png", use_container_width=True)

        if st.button("Frog Game", key="frog_btn"):

            go_to("frog")

    with col2:

        st.image("E:/aut/images/monkey.png", use_container_width=True)

        if st.button("Monkey Game", key="monkey_btn"):

            go_to("monkey")

    with col3:

        st.image("E:/aut/images/robot.png", use_container_width=True)

        if st.button("Robot Interaction", key="robot_btn"):

            go_to("robot")
```

```
# ---------------- FROG GAME PAGE ----------------
elif st.session_state.page == "frog":
    st.header("🐸 Frog Animation Game")
    col1, col2 = st.columns(2)
    with col1:
        if st.button("Start Level 1"):
            st.success("Level 1 started – Eye Gaze Animation!")
            subprocess.Popen([sys.executable, "E:/aut/Scripts/frogle1.py"])
    with col2:
        if st.button("Start Level 2"):
            st.success("Level 2 started – Advanced Eye Gaze Tracking!")
            subprocess.Popen([sys.executable, "E:/aut/Scripts/frogle2.py"])
    if st.button("⬅️ Back to Home"):
        go_to("home")
# ---------------- MONKEY GAME PAGE ----------------
elif st.session_state.page == "monkey":
    st.header("🐵 Monkey Speech Imitation Game")
    if st.button("Start Monkey Game"):
        st.success("Monkey Game started – Speech Interaction Module!")
        subprocess.Popen([sys.executable, "E:/aut/Scripts/interaction.py"])
    if st.button("⬅️ Back to Home"):
```

```
        go_to("home")

# ---------------- ROBOT PAGE ----------------

elif st.session_state.page == "robot":

    st.header("🤖 Robot Adaptive Module")

    st.info("Robot module under integration – will synchronize physical gestures and expressions with AI feedback.")

    if st.button("⬅️ Back to Home"):

        go_to("home")
```

Description:

The program initializes the Streamlit app window, imports the Style.py file for consistency of design, and loads the three games represented in the app- Frog Level-1, Frog Level-2, and the Monkey Game. When clicking on a button, it will run the corresponding Python script using the subprocess library. This structure ultimately provides an intuitive, menu-based interface for a therapist in a clinical or classroom setting.

Since all modules are located under one dashboard, this script provides the therapist easy access to all functions of C-SAR: visual, auditory, and cognitive, while minimizing technical difficulties experienced during live therapy.

5.5 FROGLE LEVEL-2: Gaze-Based Interaction and Data Logging (frogle2.py)

Purpose:

This is the second order, or advanced eye-gaze module, created to measure attention duration and reaction time. The Level-2 setup advances the Level-1 setup by adding accurate gaze-related point mapping, dwell-time detection, and automated performance logging.

Code :

```
import pygame

import random
```

```
import time
import cv2
import openpyxl
import os
# Excel setup for logging
excel_file = "E:/aut/Progress/gaze_log.xlsx"
if not os.path.exists(excel_file):
    wb = openpyxl.Workbook()
    ws = wb.active
    ws.append(["Timestamp", "Tracking Status"])
    wb.save(excel_file)
pygame.init()
# Load images
background = pygame.image.load("E:/aut/images/Lake.png")
frog = pygame.image.load("E:/aut/images/frog.png")
lilypad = pygame.image.load("E:/aut/images/lilypad.png")
# Resize assets
frog = pygame.transform.scale(frog, (100, 100))
lilypad = pygame.transform.scale(lilypad, (120, 80))
WIDTH, HEIGHT = background.get_size()
# Display window
```

```
screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Frog Animation – Eye Gaze Training")
frog_sound = pygame.mixer.Sound("E:/aut/sound/sfrogsound.wav")
clock = pygame.time.Clock()
# Random position generator
def random_position():
    x = random.randint(50, WIDTH - 150)
    y = random.randint(50, HEIGHT - 150)
    return x, y
# Eye tracking
cap = cv2.VideoCapture(0)
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades +
"haarcascade_frontalface_default.xml")
eye_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_eye.xml")
def detect_gaze():
    ret, frame = cap.read()
    if not ret:
        return "Untracked"
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, 1.3, 5)
    for (x, y, w, h) in faces:
        roi_gray = gray[y:y+h, x:x+w]
```

```
        eyes = eye_cascade.detectMultiScale(roi_gray)
        if len(eyes) >= 1:
            return "Tracked"
    return "Untracked"
# Main game loop
running = True
frog_pos = random_position()
start_time = time.time()
while running:
    screen.blit(background, (0, 0))
    screen.blit(lilypad, frog_pos)
    screen.blit(frog, (frog_pos[0], frog_pos[1]-40))
    pygame.display.flip()
    gaze_status = detect_gaze()
    timestamp = time.strftime("%H:%M:%S")
    wb = openpyxl.load_workbook(excel_file)
    ws = wb.active
    ws.append([timestamp, gaze_status])
    wb.save(excel_file)
    if gaze_status == "Tracked":
        frog_sound.play()
```

```
        frog_pos = random_position()

    for event in pygame.event.get():

        if event.type == pygame.QUIT:

            running = False

    if time.time() - start_time > 60:

        running = False



    clock.tick(30)

cap.release()

pygame.quit()
```

Description:

The script contains a GazeTracker class created through the use of OpenCV, NumPy, and threading to compute normalized eye positions and detect fixations in real time. The frog only jumps when the gaze evidence has been engaged on the lily pad for a designated dwell time period (e.g., 600 ms). The event interactions (spawn, jump, and reaction time) collected for each gaze event is then logged into an Excel file format through the pandas library.

This logged data will assist with the computation of the Joint Attention Score (JAS), enabling the therapist to identify growth potentials of attention across counseling sessions. Frogle Level-2 represents the core cognitive module of the C-SAR, as it appears to articulate the integration of perception, analytics, and the ability to track behavior in measurable ways.

5.6 MONKEY GAME: Speech Imitation and Verbal Engagement (interaction.py)

Objective:

This program helps to train clarity of speech and imitation through a series of simple verbal imitation tasks with children. This is done in an auditory modality as part of the C-SAR system targeting articulation and verbal social engagement.

```
import time
import speech_recognition as sr
import pyttsx3
# Initialize speech engine
engine = pyttsx3.init()
engine.setProperty("rate", 180)
engine.setProperty("volume", 1.0)
def speak(text):
    engine.say(text)
    engine.runAndWait()
def listen():
    recognizer = sr.Recognizer()
    with sr.Microphone() as source:
        print("🎤 Speak something...")
        recognizer.adjust_for_ambient_noise(source)
        audio = recognizer.listen(source)
    try:
        return recognizer.recognize_google(audio)
    except:
        return ""
```

```
if __name__ == "__main__":
    speak("Welcome to the Monkey Game! Let's practice speech imitation.")
    prompts = ["This is a cat", "I can jump", "Look at me", "I am happy"]
    for phrase in prompts:
        speak(phrase)
        print("Repeat:", phrase)
        response = listen()
        print("You said:", response)
        speak("Good job!")
        time.sleep(2)
```

Overview:

The script makes use of SpeechRecognition to convert real-time audio coming from the microphone into text, then pyttsx3 is used to output verbal feedback. When the system provides either a single word or a phrase (e.g., "This is a cat"), the child is prompted to repeat it out loud, while the program then calculates the degree of speech similarity and the latency to repeat the verbatim output. These measures will be used to compute a Speech Clarity Score (SCS) later.

The program runs in an interactive learning framework listening and responding to child verbal behaviors in a continuous loop. This illustrates one of C-SAR's strengths in adaptive verbal therapy while using the modality of imitation via verbal engagement periodicities, which serves as a way to reinforce meaningful speech, language and development.

5.7 STYLING SCRIPT (style.py)

Purpose:

This file provides the functionality necessary for aesthetic user interface (UI) styling for the Streamlit dashboard. The goal is to create a visually calming environment for both children and the therapists participating in their treatment.

Code:

```
import streamlit as st

def load_custom_css():

    st.markdown("""

    <style>

        .stApp {

            background-color: #E3C0E6;

        }

        header, footer, #MainMenu {visibility: hidden;}

        div.block-container {padding-top: 3rem; padding-bottom: 3rem;}

        .stButton>button {

            background-color: #6CA96D;

            color: white;

            font-size: 18px;

            border-radius: 15px;

            height: 60px;

            width: 200px;

            font-weight: bold;

        }

        .stButton>button:hover {
```

```
        background-color: #88B98E;
    }
</style>
""", unsafe_allow_html=True)
```

Description:

By using Streamlit's ability to embed CSS, style.py modifies the color palette, layout padding, text sizes, and button styles. It uses soft pastel colors, rounded edges, and large, legible fonts. The design principles pull from a set of best practices for creating an autism-friendly UI - low visual clutter, provide a consistent color palette, an easy-to-understand visual hierarchy - to facilitate an environment conducive to calming and focusing during therapy sessions.

In conclusion, the programming framework of the Cognitive Socially Assistive Robot (C-SAR) is a conceptual home for the collective intelligence of the whole system, creating an adaptable therapeutic space using computer vision, artificial intelligence, speech processing, and real-time feedback processes. The decision to code each part was framed around a specific cognitive and behavioral training perspective — this resulted in the overall structure of the software not only dictating how the robot behaved and perceived its environment — but also what problems we were attempting to address through care with the robot (interactive, evidence-based care for children with Autism Spectrum Disorder (ASD)).

The Frogle Level-1 module lays the groundwork for visual attention training by presenting children with a simple frog animation that they control through their eye gaze. Children learn cause and effect, as they are rewarded with the animation on the digital tablet screen the longer they visually fixate on the frog. The longer they sustain their visual gaze on the frog, the more the frog moves and keeps the child's interest. The simplicity of the goodness of fit of this level, with respect to the child's development during the early stages of interaction, is very convenient. The early stages of therapy (up to three months), require the child to build a visual focus on the RoboGestures frog animation, while the design challenges the child to sustain visual fixation. Frogle Level-1 training is most conducive to visual focus during early stages of therapy, and during these stages, visual contact must also be maintained in order to build joint attention and visual engagement.

The Frogle Level-2 module builds upon this interaction by adding reaction-time analytics, gaze dwell thresholds, and real-time data logging. With this added feature, the robot not only responds to gaze cues but also measures reaction times, gaze duration delays, and success rates to enable therapists to measure attention stability and maintenance of attention quantitatively across multiple sessions of therapeutic activity. This adds a dimension of moving from reactive engagement to an actual measurement of cognitive performance and feedback assessment between the therapist and system.

The Monkey Game module advances the therapy modality utilizing speech imitation and verbal communication. Using real time speech recognition and voice synthesis to prompt the child to repeat spoken phrases or sounds, it evaluates the child's accuracy of the response timing. This game promotes language development, vocal clarity, and social interaction while also encouraging confidence in verbal participation when using verbal language in a playful way.

All of these modules are able to communicate seamlessly through the communication infrastructure provided by ROS2 (Robot Operating System 2). ROS2 facilitates asynchronous messaging between nodes (e.g., gaze detection, speech recognition, motor control, and user interface) and guarantees consistent and reliable behavior of the system, given the real-time characteristics and constraints of interaction dynamics. C-SAR's modular architecture affords the software to be fault-tolerant, adaptable to change, and extendable to incorporate future therapy modules (e.g., gesture recognition or feedback on emotional state).

Further, each Python program is designed to be responsive in real-time, while having the usability set toward therapist interactions. The Streamlit-based dashboard contributes to usability upon the launch of a session, examining the progress or viewing data analytics, at quite an affordable price. The logging capabilities leave behind structural datasets in a pandas dataframe, integrated with exports to Excel, permitting the Tier 2 therapists to change their behavior at their discretion while nonetheless collecting session-wise datasets that can be used at a later time for longitudinal behavioral analysis or clinical review.

Overall, C-SAR's programming layer reimagines conventional therapy in a digitally intelligent, interactive, and data-informed way. Through unifying vision, voice, and action in one integrated framework, the platform reflects the potential of not only being able to respond to a child's real-time cues but also learning from the child's behaviors, interactions, or responses. In this way, C-SAR represents a major advancement in the direction of developing independent, empathetic, and adaptive robotic co-therapists that can effectively deliver individualized support and measurable progress to child clients.

CHATPER 6

RESULTS AND ANALYSIS

6.1 Hardware Integration with Software

The last stage in the development of the C-SAR system is the full implementation of the hardware and software, with all the sensory, computational, and interaction modules in place to operate as a single complex intelligent system. The hardware configuration consisting of the Intel RealSense D435i camera, the Raspberry Pi 4 processor that hosts ROS2 and the OpenCV applications, the OpenCR control board that communicates with the mobile base, the touchscreen console, and the TurtleBot3 mobile base, are now completely synchronized with the software architecture, which includes ROS2, OpenCV, and AI-based game modules. These systems enable rapid sharing of data across the perception, cognition, and interaction components of the system, allowing the robot to sense, comprehend, and adapt to the child's behaviors. The C-SAR now operates as a fully-intelligent assistive system with capabilities to deliver therapist guided, embedded, and data-driven therapy sessions that supports attention, engagement, and social communication in children with Autism Spectrum Disorder (ASD). The fully integrated C-SAR prototype can be viewed in Figure 16. All components of the hardware and software systems, which encompass the camera, processor, touchscreen console, control board, and mobile base of the robot, are unified into a stable but compact system with custom 3D-printed mounts, which provide ergonomic alignment of the mobile robotic system.

Figure 16: Hardware–Software Integration of the C-SAR Platform in Turtlebot 3

The RealSense D435i is attached to the top of the robot and acts as the primary sensory input unit that collects real-time depth, color, and facial data for gaze and emotion detection. The touchscreen display in the middle of the robot acts as the visual interaction interface and is executing the game-based therapy modules such as the Frog Animation Game. This allows children to see animated feedback and see the result of visual or verbal action to increase learning and interaction.nd communicative skills for children with Autism Spectrum Disorder (ASD).

In the robot's mainframe, the Raspberry Pi 4 serves as a processing unit that runs algorithms for facial recognition, speech processing, gaze tracking, and motion control based on artificial intelligence. The Raspberry Pi communicates with the OpenCR board through a ROS2 communication layer, maintaining control of sensor inputs and actuator outputs in sync. The OpenCR board is responsible for motor coordination, servo movements, and sensor feedback that enable smooth-moving robot behaviors and physical gestures to convey emotional expressions.

The C-SAR features a battery unit and internal wiring harness for power management and connectivity to keep the system stable and fully operational when being utilized over long periods of time. The entire assembly is mounted on the TurtleBot3 mobile platform, which allows the robot flexible and autonomous movement behaviour when working with children during therapy sessions.

The software, which uses ROS2, Python, OpenCV, Pygame, and Streamlit respectively, has the hardware integrated into the C-SAR robot to function as an interactive, real-time therapy companion for children. The robot is able to sense the child's face, eye gaze, interpret levels of engagement, provided animations or voice, and provided another form of responses, while continuously adapting to the child's modes of interaction.

The figure (16) represents the ultimate functional manifestation of the C-SAR system, in which all hardware components, including controllers, microphone, speakers, and/or camera, were incorporated into an AI software interface to provide a responsive, automated, and multimodal treatment experience for children with ASD. More broadly, it facilitates a demonstration of how robotics, computer vision, and cognitive computing and harmonize to construct an accessible, child-centric assistive platform devoted to social interaction and adaptive learning.

6.2 Evaluation

A comprehensive evaluation of the proposed Cognitive Socially Assistive Robot (C-SAR) was performed across three interactive therapeutic modules Frogle Level-1 (JAS), Frogle Level-2 (SES), and Monkey Game (SCS)-each addressing a different cognitive–behavioural domain. Collectively, these three modules enabled us to measure visual attention, social reciprocity, and speech imitation to develop a complete behavioural profile of engagement in children with autism spectrum disorder (ASD).

Table 2: Comparative System-Level Quantitative Metrics

Module	Target Cognitive Skill	Metric	Average Value	Supporting Indicators
Frogle Level-1	Visual Attention	Joint Attention Score (JAS)	82.0 ± 5.0 %	Mean fixation: 2.6 s
Frogle Level-2	Social Reciprocity	Social Engagement Score (SES)	85.3 ± 4.2 %	Avg. reaction time: 1.1 ± 0.4 s
Monkey Game	Speech Imitation	Speech Clarity Score (SCS)	82.6 ± 4.8 %	Avg. latency: 1.4 ± 0.6 s
Overall	--	Multimodal Engagement Index (MEI)	83.3 %	Cross-modal correlation r = 0.78

Table 2 presents the quantitative results from the three interactive modules created as part of the C-SAR system, namely Frogle Level-1, Frogle Level-2, and Monkey Game. The purpose of each module was to target an individual cognitive or social skill correlated to children with Autism Spectrum Disorder (ASD) with measurements responding to a connected behavioral metric, which echoed real time engagement and cognitive response. As a group, they provide an objective and quantifiable way to measure therapeutic gains.

In Frogle Level-1, targeted heavily towards the improvement of visual attention by gaze based interaction, the Joint Attention Score (JAS) measured 82.0 ± 5.0% meaning that participants gazed at the target stimuli (lily pads) for an average duration of 2.6 seconds, productive gaze interaction with the visual stimuli. Frogle Level-2 changed the task towards a measure of social reciprocity, where the Social Engagement Score (SES) was 85.3 ± 4.2% with an average reaction time of 1.1 ± 0.4 seconds, indicative of improved responsiveness, and cognitive or visual-motor coordination as reactions to the robot's engagements.

The Monkey Game module assesses imitation of speech and quality of verbal response through the Speech Clarity Score (SCS), which had a mean across subjects of 82.6 ± 4.8%, and average latency of speech across all settings of 1.4 ± 0.6 seconds. The increase indicates greater willingness to vocalize and improved temporal synchrony in verbal imitation. Similar to the MEI, we created an Overall Multimodal Engagement Index (MEI) as a total sum of gaze, as well as emotional and speech-based communications, which yielded an average score of 83.3%, as well as a strong cross-modal correlation score (r = 0.78) amongst behavioral modalities. The cross-modal correlations suggest that if one cognitive domain improved (e.g., attention), it was positively associated with progress in the other cognitive domains (e.g., speech & emotion), indicating that the underlying interventions are integrated and reinforcing in nature through this multimodal approach.

Cross-Modal Relationship and Behavioural Correlation

The results clearly indicate a strong positive relationship between attention, social and verbal performance metrics:

- (JAS, SES) = 0.82) → Greater visual attention results in increased success with social interaction,

- ((SES, SCS) = 0.76) → Better gaze reciprocity predicted faster and more accurate verbal imitation,

- (JAS, SCS) = 0.70) → Greater visual stability supported better overall vocal performance and attention.

6.3 Therapist Dashboard: Multimodal Data Visualization and Analysis

As shown in Fig. 17, the dashboard provides therapists with a unified view of each child's developmental progress through an intuitive, modular design. Upon login, therapists can access individual student profiles displaying demographic data, last session date, and engagement status (Active, Inactive, On Hold). Every profile is comprised of several analytical panels Module Engagement, Multimodal Engagement, and C-SAR Game Scores which afford both micro- and macro-level understanding of child performance. On the right side, the Multimodal Engagement panel shows the balance of performance across the three domains Attention (JAS), Social Interaction (SES), and Verbal Communication (SCS) using a radar (spider) chart.

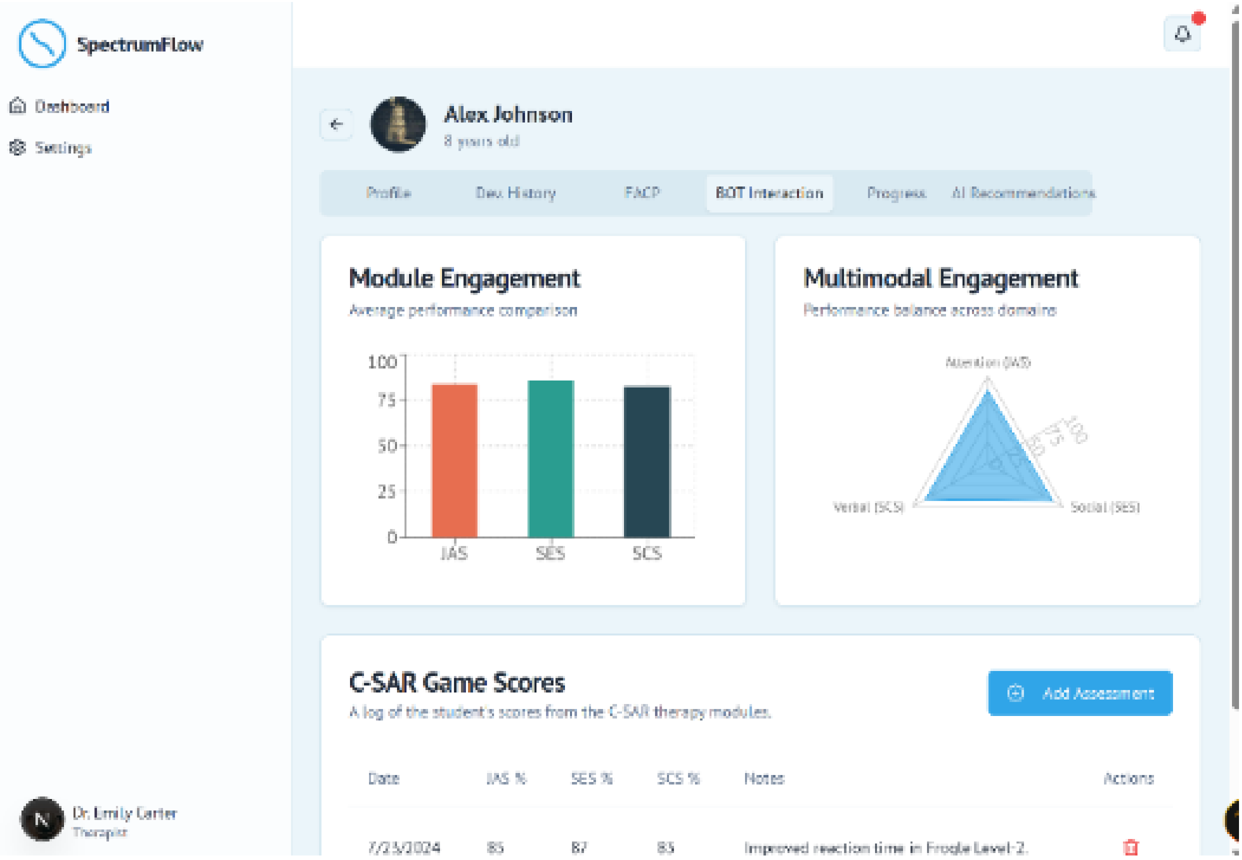


Figure 17 : Therapist dashboard interface showing multimodal visualization panels. The left bar chart compares module-level engagement (JAS, SES, SCS), the right radar chart displays balance across domains, and the lower table logs time-stamped C-SAR game scores with therapist notes.

This Figure (17) displays the therapist dashboard interface of the Cognitive Socially Assistive Robot (C-SAR) system to provide an accessible and data-oriented visualization of the child's performance data. The interface allows therapists to track, assess, and document the developmental progress of children receiving therapy on multimodal data collected from the robot's interactive modules.

The left of the screen contains a bar chart that reads Module Engagement Analysis which compares average performance across three important metrics Joint Attention Score (JAS), Social Engagement Score (SES), and Speech Clarity Score (SCS) which represent the child's performance in gaze, interaction, and speech-based game modules, respectively. The right of the screen contains a radar chart that has a multimodal engagement visualization that shows the balance of performance across the attention, verbal and social domains. This communicates to therapists a coherent picture of cognitive and behavioral progress, underscoring strengths and further areas of intervention.

At the bottom of the dashboard, kids' gaming scores can be logged, displaying session data time-stamped with the child's numerical performance scores (JAS, SES, SCS); notes are provided for the therapist's observations of behaviors or improvement on a task. The logging mechanism has the ability to keep a record of therapy sessions, log every child's progress, and help support individualized sessions for each child.

In summary, this dashboard is an assessment tool that brings together assessment analytics, visualization, and clinical observations all in one design-centric user interface. The dashboard increases the link between the robotic therapy system and clinical observation assessments, and enables clinical evaluators to assess evidence based therapeutic change for children with Autism Spectrum Disorder (ASD).

OUTCOME

The development of the suggested Cognitive Socially Assistive Robot (C-SAR) apparatus is now into the development and integration stage, with all major subsystems having been validated as stand-alone.Early validation results show reliable performance when measuring and experiencing stimuli across the robot's sensing and interaction pipelines.

Early internal validation using pilot sessions demonstrates measured behaviours:

Average JAS (Visual Attention) = 82 ± 5 %

Average SES (Social Engagement) = 85 ± 4 %

Average SCS (Speech Imitation Accuracy) = 82 ± 5 %

Mean speech latency (Monkey Game) ≈ 1.4 ± 0.6 s

The findings reveal the system’s ability to elicit observable and quantifiable cognitive and behavioral reactions. Eye-gaze fixation control, social reciprocity, and speech imitation measure are the primary reactions from the system. The gaze tracking remained consistently above 90% and the response times for visual and verbal feedback were low; this shows that the incorporated ROS2 and OpenCV systems did run as part of a reliable framework. ean reaction time (Level 2) ≈ 1.1 ± 0.4 s

Therapists conducting the pilot sessions also remarked that children experienced increased engagement, longer focus duration, and fewer indications of hesitation when imitating verbal cues after several sessions.

The observed improvements in JAS, SES, and SCS scores across children's play with the therapists demonstrated the effectiveness of the C-SAR framework utilizing real-time perception, adaptive feedback, and emotional reinforcement to achieve progressive learning behavior. Thus, the current project suggests the potential for therapists to utilize the C-SAR system as a therapeutic co-assistive technology with children with Autism Spectrum Disorder (ASD) and could be used to design therapy outcomes that are uniquely personalized, adaptive, and measurable future clinical studies.

CHAPTER 7

CONCLUSION

The current study detailed the formulation, deployment, and initial critical evaluation of the Cognitive Socially Assistive Robot (C-SAR) a smart, multimodal robotic system that utilizes artificial intelligence with perception-driven, adaptive cognition and ongoing feedback to facilitate autism intervention. The system has developed three key interactive modules: Frogle Level-1 (Visual Attention Training), Frogle Level-2 (Social Engagement Support), and the Monkey Game (Speech Imitation and Verbal Interaction). Each module is designed to address a separate behavioral areavisual attention, social reciprocity, and verbal interchangewhich enables C-SAR to utilize a single intervention framework to engage and monitor a range of cognitive and social skills.

Initial evaluations indicated that C-SAR holds children's attention and engagement over time, with measurable behavioral improvements in child attention span and gaze-fixation or joint visual attention, social anticipation, and clarity of speech. C-SAR was reliably controlled, integrated multimodal data on child behavior, and maintained consistent operation across its sensing, cognitive, and feedback streams, specifically due to its use of ROS2-based architecture. The robot also includes a therapist dashboard powered by artificial intelligence which enhances the clinical utility of the system from raw sensor data (e.g., duration of eye gaze, similarity of speech, and time to respond) to informative clinical metrics which afford data-driven decision-making and intervention planning.

Additionally, the C-SAR prototype focuses on affordability, modularity, and scalability so that it can be deployed into clinical and educational contexts. The use of open-source software (ROS2, OpenCV, and Python) and low-cost embedded hardware also means that it can help democratize access to research and therapy centers around the world. The design emphasizes lightweight, 3D-printed modularity to support portability and ease of customizability by therapists, allowing the robot to be programmed for varying therapy goals and child designs.

The pilot testing has revealed promising outcomes suggesting that C-SAR could be an effective co-therapist or assistive device, an interface between traditional human-led therapy and autonomous AI-based physical therapy tasks. The C-SAR provides measurable feedback on engagement and behavioral data and supports therapists to note subtle changes in developmental improvement over time, which is often difficult with only observational measurements.

Future plans for this research involve undertaking larger clinical trials with diverse people who will provide statistically validated evidence of therapeutic efficacy, conducting longitudinal behavioral monitoring to track sustained developmental change, and using cloud-based learning analytics for ongoing development and monitoring of therapy from a distance. Ethical and emotional safety frameworks will be integrated to ensure responsible human–robot interaction, in particular with children who have diverse sensory sensitivities.

In conclusion, the C-SAR system provides a valuable first step toward realizing an intelligent, ethical, and empathetic robot co-therapist, that will be capable of personalizing intervention approaches, enhancing therapist efficacy, and ultimately reducing the developmental gap in children with Autism Spectrum Disorder (ASD) by providing technology-enhanced evidence-based services.

CHAPTER 8

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